Head-to-Tail Interference in Long-Tailed Small-Batch Training

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Abstract

Long-tailed small-batch training creates a local stability problem: a head-class update can improve the current mini-batch while perturbing logits on rare tail examples that are absent from the update. We study this problem through a matched-head-gain diagnostic: after two update directions make the same first-order progress on the head batch, compare their squared logit drift on held-out tail examples. For a sandwich matrix block $J_T(D) = B_T D A_T$, the resulting interference coefficient predicts a smaller worst-case drift bound for spectral/polar geometry when $\mathrm{nrank}(G_H) > \mathrm{ssrank}(B_T, A_T)$, where $\mathrm{ssrank}(B_T, A_T)$ is a downstream-aware stable-rank quantity. We make four contributions: a matched-head-gain formulation, the sandwich-block condition, an arbitrary-direction protocol for Muon-style directions, and controlled diagnostics that test the claim boundary. Controlled boundary experiments reverse the spectral/Frobenius squared drift ratio from 0.3403 to 7.208, a controlled long-tailed digits diagnostic gives a spectral/Frobenius squared tail-example logit drift ratio of 0.5501 [0.5101, 0.5931], and a GPU CIFAR-100-LT ResNet18 architecture check gives 0.5611 [0.5224, 0.6026]. Tail-rich and all-layer ResNet JVP controls further test whether the local signal survives a more meaningful tail function and earlier convolutional blocks. The result is a local function-drift mechanism, not an optimizer-level performance claim: lower logit drift does not automatically improve tail cross-entropy, margin, or accuracy.

1 Introduction

Long-tailed small-batch training repeatedly updates a model on examples that are not representative of the full function we would like to maintain. Frequent head classes dominate many mini-batches, while rare tail classes may be absent for several consecutive steps. During those gaps, a head-only update can reduce the current batch loss while changing the logits of held-out tail examples. This creates a local function-drift problem: an update can make progress on the batch that produced it while perturbing examples that did not participate in the update.

The local question studied in this paper is simple: after matching the same head-batch gain, which update perturbs the tail function less?

Raw update comparisons do not answer this question because a direction can look stable simply by taking a smaller effective step. We therefore compare updates under a matched-progress rule: *match head progress, then compare tail drift*. Given a head-batch gradient g_H, we fix the same first-order head gain $-\langle g_H, \Delta\rangle = \rho$, and then measure the induced tail-output drift $\|J_T \Delta\|$. This matched-head-gain protocol turns an optimizer comparison into a local geometric question.

Spectral updates are a natural test case for this local question because they change the geometry of steepest descent. Frobenius/GD follows the normalized gradient, while spectral/polar updates emphasize the polar factor of a matrix gradient and are closely related to the matrix direction used by Muon-style optimizers. This suggests asking whether spectral geometry only changes same-batch descent, or whether it also changes how head updates perturb held-out tail functions.

The main result identifies when spectral/polar geometry has a smaller worst-case matched-gain tail-drift bound than Frobenius/GD geometry. For a matrix block whose tail perturbation has the sandwich form $J_T(D) = B_T D A_T$, the comparison reduces to the rank condition $\mathrm{nrank}(G_H) > \mathrm{ssrank}(B_T, A_T)$. The left-hand side is the nuclear rank of the head-batch gradient, namely the squared nuclear-to-Frobenius norm ratio. The right-hand side is a downstream-aware tail sensitivity rank built from the singular spectra of the tail activation and downstream Jacobian. The condition therefore depends on both the spectral spread of the head gradient and how the tail function reads the perturbed layer.

We make four contributions:

1. We formulate a local head-to-tail interference problem: after matching the same first-order head-batch gain, compare how much different update geometries perturb a held-out tail function.
2. We define a head-to-tail interference coefficient and prove a matched-gain drift bound for any norm geometry.
3. For a matrix block $J_T(D) = B_T D A_T$, we derive the spectral-vs-Frobenius condition $\mathrm{nrank}(G_H) > \mathrm{ssrank}(B_T, A_T)$.
4. We provide empirical diagnostics on a synthetic boundary case, a controlled long-tailed digits MLP, and CIFAR-100-LT ResNet18 architecture checks, including selected-state compatibility checks for Muon-style momentum and Newton–Schulz directions.

The experiments are diagnostic rather than benchmark-driven. They test whether the local drift mechanism appears in controlled matrix blocks, in a controlled long-tailed digits MLP, in CIFAR-100-LT ResNet18 one-step and layerwise architecture checks, and along short Muon-style direction trajectories. The main empirical signal is the matched-head-gain squared tail-example logit drift ratio. The synthetic positive and negative boundary constructions give squared drift ratios 0.3403 and 7.208; the controlled long-tailed digits one-step diagnostic gives squared drift ratio 0.5501 [0.5101, 0.5931]; the CIFAR-100-LT ResNet18 check gives 0.5611 [0.5224, 0.6026]; the all-layer ResNet JVP control gives observed squared drift ratio 0.2011 [0.1845, 0.2192]; the short digits Muon-style trajectory gives $\mathrm{NS}(M_t)$ squared drift ratio 0.8019 [0.7583, 0.848]; and the ResNet practical bridge gives $\mathrm{NS}(M_t)$ squared drift ratios 0.8628 [0.8154, 0.913] on AdamW-sampled states and 0.7247 [0.676, 0.777] on NS-Muon-sampled states. Accordingly, tail loss, margin, and accuracy are reported separately to avoid conflating local function-drift reduction with final classification performance.

The supported mechanism is also narrower than "spectral directions are always safer". The layerwise diagnostic shows that polar unit directions can have larger tail JVPs, while matched-head-gain scaling can still reduce observed drift. Thus the empirical claim is about tail drift after head progress has been matched, not about uniformly lower tail sensitivity of every spectral unit direction.

2 Related Work

2.1 Local Geometry of Spectral Gradients and Muon

Spectral-gradient theory provides the local geometry that motivates this paper, but it usually studies a same-batch objective. Davis & Drusvyatskiy (2025) give the layerwise condition $\mathrm{nrank}(G) > \mathrm{srank}(A)$, under which a spectral update can obtain a larger local decrease than a Frobenius update. Other recent work analyzes Muon through spectral-norm constraints (Chen et al., 2025), studies spectral anisotropy or spectral clipping in LLM training (Huang et al., 2026; Jiang et al., 2026), and examines Muon recipe interactions and gradient spectra in vision transformers (Southworth et al., 2026). These works motivate the idea that update geometry can matter beyond scalar learning-rate choice.

We use the same geometric distinction for a different local question. Rather than asking whether a spectral update decreases the head-batch loss more, we ask whether it perturbs a held-out tail function less after the head-batch gain has been matched. The resulting condition, $\mathrm{nrank}(G_H) > \mathrm{ssrank}(B_T, A_T)$, is cross-distribution: the left-hand side comes from the head gradient, while the right-hand side comes from tail activations and the downstream Jacobian.

2.2 Long-Tailed Learning and Optimizer Geometry

Classical long-tailed recognition methods address imbalance through sampling, losses, margins, or classifier calibration, including class-balanced loss (Cui et al., 2019), LDAM (Cao et al., 2019), OLTR (Liu et al., 2019), and decoupled training (Kang et al., 2020). These methods primarily change the training objective, data distribution, or classifier calibration so that rare classes receive more effective supervision. Our diagnostic studies a different failure mode: even before the next tail sample appears, a head-only update can move the model outputs on held-out tail examples.

Recent work has also connected Muon-style optimization to long-tailed recognition (Luo et al., 2025) and tail-end associative memory learning (Wang et al., 2025). This paper is complementary to those lines, but it asks a narrower mechanism question. It does not introduce a re-weighting method, and it does not claim a full long-tailed benchmark improvement. Instead, it isolates a local quantity that can be measured at a fixed checkpoint: when tail samples are absent, how much do head-only updates perturb logits on held-out tail examples?

2.3 Scope and Relation to Full Muon

The theory analyzes an idealized spectral-gradient/polar direction, while practical Muon contains additional optimizer state. Momentum, finite Newton–Schulz iterations, weight decay, schedules, and non-matrix parameter handling can all change the realized update. The theoretical conclusion should therefore be read as a local geometric mechanism: when $\operatorname{nrank}(G_H) > \operatorname{ssrank}(B_T, A_T)$ holds, spectral geometry has a smaller worst-case matched-gain upper bound on tail drift. The Muon experiments in this paper test whether selected Muon-style directions exhibit squared drift ratios compatible with this local effect; they do not constitute a theory of complete Muon training.

3 Problem Setup

We first formalize the local comparison that all theory and diagnostics use. Basic matrix-norm and rank definitions are collected in Appendix A.

3.1 Head and Tail Batches

The local comparison separates the batch that drives the update from the examples whose function we monitor. Let H denote the current head batch and let $\mathcal{L}_H(\theta)$ be its loss, with gradient $g_H = \nabla_\theta \mathcal{L}_H(\theta)$. Let T denote a tail probe batch that is not used to form the update, and let $F_T(\theta)$ collect the model outputs on T. For a small parameter perturbation Δ, the local tail-function change is approximated by $J_T(\theta)\Delta$, where $J_T(\theta) = D_\theta F_T(\theta)$. This quantity is not a training objective; it is the first-order change in tail outputs induced by an update chosen from the head batch.

The protocol fixes head progress before measuring tail drift. The first-order head gain of Δ is $-\langle g_H, \Delta\rangle$, and the tail drift is $\|J_T\Delta\|$. Rather than comparing raw step sizes or raw loss decreases, we restrict attention to updates with the same head gain ρ and ask which update geometry yields smaller tail drift. This is the object analyzed below.

3.2 Local Comparison Protocol

Assumption 3.1 (Local head-to-tail comparison protocol)**.** The one-step comparison is local: all quantities are evaluated at the same parameter θ, head batch H, and tail probe batch T. We assume:

1. $\mathcal{L}_H$ is differentiable at θ, and $g_H = \nabla_\theta \mathcal{L}_H(\theta) \neq 0$.
2. F_T admits the first-order approximation $F_T(\theta + \Delta) - F_T(\theta) \approx J_T(\theta)\Delta$.
3. The norm-N unit ball is compact, and the step is small enough for $-\langle g_H, \Delta\rangle$ to be the matched-head-gain scale.

These assumptions define a one-step diagnostic, not a complete training model. They do not assume that a full trajectory has a fixed Jacobian, nor that tail loss or accuracy is determined by logit drift alone. The experiments therefore report observed drift, tail loss, margin, and caveats separately.

4 THEORETICAL ANALYSIS

4.1 HEAD-TO-TAIL INTERFERENCE COEFFICIENT

The matched-gain protocol reduces the comparison of two update geometries to a sensitivity-per-progress quantity. Let $\|\cdot\|_N$ be a parameter norm and $\|\cdot\|_{N,*}$ its dual norm. We define the unit-step tail sensitivity $K_{T,N} = \sup_{\|\Delta\|_N \le 1} \|J_T \Delta\|$ and normalize it by the dual norm of the head gradient:

$$\boxed{\mathcal{I}_N(T \mid H) = \frac{K_{T,N}^2}{\|g_H\|_{N,*}^2}.}$$

This coefficient measures the worst-case squared tail-output drift upper bound per unit squared first-order head gain under norm-N optimization geometry.

4.2 GENERAL MATCHED-GAIN DRIFT BOUND

Theorem 4.1 (Tail drift under matched head gain)**.** *Under Assumption 3.1, let u_N be a norm-N steepest direction and set*

$$\Delta_N = -\frac{\rho}{\|g_H\|_{N,*}} u_N.$$

Then $-\langle g_H, \Delta_N \rangle = \rho$ and

$$\|J_T \Delta_N\|^2 \le \rho^2 \mathcal{I}_N(T \mid H).$$

Therefore, among two update geometries with the same matched head gain, the geometry with the smaller $\mathcal{I}_N(T \mid H)$ has the smaller worst-case tail-drift bound.

The proof is a direct dual-norm scaling argument and is included in Appendix B. The important point is the comparison protocol: raw progress is not enough; drift is compared only after the same first-order head gain has been fixed.

Worst-case and realized coefficients. The coefficient $\mathcal{I}_N$ is a worst-case sensitivity over the whole norm unit ball. The experiments do not estimate this supremum. They estimate the realized actual-direction coefficient

$$\widetilde{\mathcal{I}}_N(T \mid H) = \frac{\|J_T u_N\|^2}{\|g_H\|_{N,*}^2} \le \mathcal{I}_N(T \mid H),$$

where u_N is the update direction actually used. Thus the theorem compares worst-case sensitivity-per-progress, while observed drift also depends on alignment between u_N and the tail Jacobian. The synthetic boundary experiment is designed to test when the worst-case comparison is directionally informative; the neural diagnostics report realized drift rather than claiming equality with the bound.

Arbitrary-direction matched gain. Muon-style momentum directions require a slightly more general normalization because $\mathrm{polar}(M_t)$ and finite $\mathrm{NS}(M_t)$ are not steepest directions for the current gradient g_H. For any descent direction d satisfying $\langle g_H, d \rangle > 0$, we use the matched-gain step

$$\Delta_d = -\rho \frac{d}{\langle g_H, d \rangle}, \qquad C(d; T \mid H) = \frac{\|J_T d\|^2}{\langle g_H, d \rangle^2}.$$

The measured drift is then $\|J_T \Delta_d\|^2 = \rho^2 C(d; T \mid H)$. When d is a norm-steepest unit direction, $\langle g_H, d \rangle = \|g_H\|_{N,*}$, and this reduces to the realized coefficient above. For $\mathrm{polar}(M_t)$ and $\mathrm{NS}(M_t)$, all reported Muon-style ratios use this actual head-alignment denominator $\langle g_H, d \rangle$, not the dual norm of the current gradient or momentum state.

4.3 MATRIX-BLOCK SPECTRAL CONDITION

The general coefficient becomes concrete for a matrix parameter block W. Let $G_H = \nabla_W \mathcal{L}_H$ be the head-batch matrix gradient. Assume the local effect of perturbing this block on the tail outputs has the sandwich form $J_T(D) = B_T D A_T$, where A_T is the tail activation entering the layer, B_T is the downstream tail Jacobian, and $D = \Delta W$ is the block perturbation. This assumption is exact for linear readout blocks, final linear layers, or shared downstream linearizations; for general nonlinear blocks, the same coefficient definition applies but the closed-form sandwich rank may not.

For this sandwich block, Appendix C derives the Frobenius and spectral interference coefficients. Comparing them gives the paper's central condition:

$$\boxed{\operatorname{nrank}(G_H) > \operatorname{ssrank}(B_T, A_T).}$$

The downstream-aware stable-rank term is

$$\operatorname{ssrank}(B_T, A_T) = \frac{\sum_i \sigma_i(B_T)^2 \sigma_i(A_T)^2}{\sigma_1(B_T)^2 \sigma_1(A_T)^2},$$

with singular values sorted in descending order and padded by zeros after the shorter sequence. Equivalently, the bound ratio is

$$\frac{\mathcal{I}_S(T \mid H)}{\mathcal{I}_F(T \mid H)} = \frac{\operatorname{ssrank}(B_T, A_T)}{\operatorname{nrank}(G_H)}.$$

Here $\operatorname{nrank}(G_H)$ measures how spectrally spread the head gradient is, while $\operatorname{ssrank}(B_T, A_T)$ is not merely the stable rank of tail activations: it pairs the singular spectrum of the downstream tail Jacobian with the singular spectrum of the tail activation matrix. The condition says that the spectral matched-gain drift bound is smaller when the head gradient is spread enough to offset downstream tail sensitivity. It is still a worst-case sensitivity upper bound: the observed drift of $u_S = \operatorname{polar}(G_H)$ can also depend on the alignment between the singular vectors of G_H and the tail Jacobian. The experiments therefore report both condition-side quantities and directly measured tail-output drift.

5 Experiments

5.1 Experiment Protocol Summary

The experiments test the proposed mechanism through a sequence of increasingly realistic diagnostics. The synthetic matrix block asks whether the rank boundary can reverse the drift ordering. The controlled long-tailed digits diagnostic asks whether the same matched-head-gain drift effect appears in a trained neural model. The CIFAR-100-LT ResNet18 diagnostics ask whether the one-step readout survives a convolutional architecture with BatchNorm state fixed during measurement, whether it persists at a tail-rich checkpoint, and whether all Conv/Linear matrix weights show consistent finite-difference JVP and observed layer-only drift behavior. The Muon-style direction compatibility diagnostic asks whether selected momentum and Newton–Schulz directions have squared drift ratios compatible with the local polar mechanism. The layerwise diagnostic asks whether lower drift comes from intrinsically safer unit directions or from matched-head-gain scaling.

All experiments use the same comparison principle unless stated otherwise: construct Frobenius/GD and spectral/polar/Muon-style directions on a head batch, scale them to the same first-order head gain ρ, and then measure held-out squared tail-example logit drift. Unless otherwise stated, ratios below are squared drift ratios, spectral/polar divided by Frobenius/GD; values below 1 mean less squared logit drift on held-out tail examples under matched head gain. Split, seed, batch, noise, warmup, finite-difference, and confidence-interval details are in Appendix I. These intervals quantify paired uncertainty under the tested distribution; they are not generalization guarantees.

Throughout the experiments, tail-example logit drift means the drift of the full class-logit vector evaluated on held-out tail examples. It is not restricted to logits of tail classes only; head-class logits on tail examples are included because they affect cross-entropy, margins, and predictions.

For the two-layer MLP diagnostics, the compared directions are blockwise matrix directions over the two weight matrices W_1, W_2. The model has no bias, batch-normalization, or non-matrix parameters, so there are no classifier-bias or non-matrix update rules to specify. Frobenius/GD uses the globally Frobenius-normalized full gradient $d_F = \{G_\ell\}_\ell / \sqrt{\sum_\ell \|G_\ell\|_F^2}$. Spectral/polar uses the blockwise direction $d_S = \{\operatorname{polar}(G_\ell)\}_{\ell=1}^2$, which is the steepest direction for a max-over-block spectral norm as noted in Appendix G. Muon-style directions replace G_ℓ by the momentum state M_ℓ and optionally apply finite Newton–Schulz iterations. All these directions are finally scaled by the same arbitrary-direction rule $\Delta_d = -\rho d / \langle g_H, d\rangle$, so the comparison is a blockwise geometry diagnostic rather than a single global spectral-norm theorem for complete training. Rank and activation diagnostics are computed on the full held-out tail probe set, while the update direction itself is formed from the head mini-batch.

Because the theory is first-order, we also check the local linearization quality at the controlled long-tailed digits checkpoint. Appendix Table 5 reports $\|F_T(\theta+\Delta) - F_T(\theta) - J_T\Delta\| \,/\, \|J_T\Delta\|$ for Fro/GD, $\mathrm{polar}(G_t)$, $\mathrm{polar}(M_t)$, and $\mathrm{NS}(M_t)$; the largest reported 95% upper endpoint is 2.05×10^{-3}, attained by $\mathrm{polar}(M_t)$. Thus the matched-gain measurements in this diagnostic are not dominated by finite-step nonlinear drift at the chosen step scale.

5.2 Synthetic Head-to-Tail Linear Model

The synthetic experiment tests whether the theorem's boundary has operational content. We isolate the condition $\mathrm{nrank}(G_H) > \mathrm{ssrank}(B_T, A_T)$ from modeling complications by constructing a linear sandwich tail map $F_T(W) = B_T W A_T$, so that the local perturbation is $J_T(D) = B_T D A_T$, and controlling the singular values and aligned singular-vector geometry of G_H, A_T, B_T. After matching first-order head gain, we measure the spectral/Frobenius squared tail-output drift ratio. The random seeds in this diagnostic regenerate the controlled construction but do not introduce independent variation in the aligned spectra, so the confidence intervals collapse to point intervals; this should be read as a deterministic boundary check, not as a stochastic generalization experiment.

The drift ordering changes with the boundary. In the positive construction, $\mathrm{nrank}(G_H) = 8.607$ exceeds $\mathrm{ssrank}(B_T, A_T) = 1$, and the observed spectral/Frobenius squared tail-output drift ratio is 0.3403 [0.3403, 0.3403]. In the negative construction, $\mathrm{nrank}(G_H) = 1.387$ is below $\mathrm{ssrank}(B_T, A_T) = 10$, and the observed squared drift ratio becomes 7.208 [7.208, 7.208]. This sign change is the key boundary check: $\mathrm{ssrank}(B_T, A_T)$ is not merely notation, but the tail-side quantity that determines when the sandwich worst-case bound favors spectral geometry.

The same synthetic setup also shows why the condition should not be overread as a realized-drift predictor by itself. Appendix Figure 4 keeps the same positive singular spectra, so the bound ratio remains 0.1162, but randomizes the singular-vector alignment among G_H, A_T, B_T. Under this randomized alignment, the observed squared drift ratio is 1.171 [1.069, 1.283], with median 1.303, and spectral has lower drift in only 0.416 of 500 trials. The rank condition is therefore a bound-ordering condition; realized drift also depends on alignment with the tail perturbation operator.

5.3 One-Step Diagnostic on Controlled Long-Tailed Digits

The controlled long-tailed digits diagnostic asks whether the same drift reduction appears away from the controlled linear block. We use the scikit-learn 8×8 digits image dataset (Pedregosa et al., 2011), scale pixel values to the unit interval, fix classes 0–4 as head classes and classes 5–9 as tail classes, and sample the per-class train/evaluation split independently for each seed. A bias-free two-layer MLP is trained to a head-heavy checkpoint; a head-only batch defines the update and a held-out tail set measures drift. This is a mechanism probe, not a standard long-tailed benchmark.

In the one-step diagnostic, spectral/polar reduces squared tail-example logit drift in every tested seed (20 total). The paired squared drift ratio is 0.5501 [0.5101, 0.5931]. Centering each tail-example logit vector gives almost the same ratio, 0.5493 [0.5093, 0.5923], so the effect is not only a common-mode logit shift. The margin-relevant components are more mixed: the top-competitor logit-delta ratio is 0.6358 [0.5705, 0.7086], the margin-delta ratio is 0.702 [0.6131, 0.8037], but the true-class logit-delta ratio is 1.68 [1.16, 2.431]. This supports a drift claim while keeping the CE/margin claim narrow. Appendix Figures 7, 8, 9, and 10 stress-test this readout across 4 tail-sample levels, 5 warmup checkpoints, 5 head/tail class partitions, and 5 target head-gain fractions. The largest squared drift-ratio upper endpoints are 0.6361 across imbalance settings, 0.7071 across checkpoint settings, 0.6532 across class partitions, and 0.5932 across target head-gain fractions, so the lower-drift readout is not restricted to the default 100:40, 80-step, low-vs-high digit state or to the default $\rho = 0.02 L_H$ scale. However, this checkpoint sweep still does not show preservation of a high-quality tail predictor: across the tested checkpoints, pre-update tail accuracy spans only 0 to 0.192,

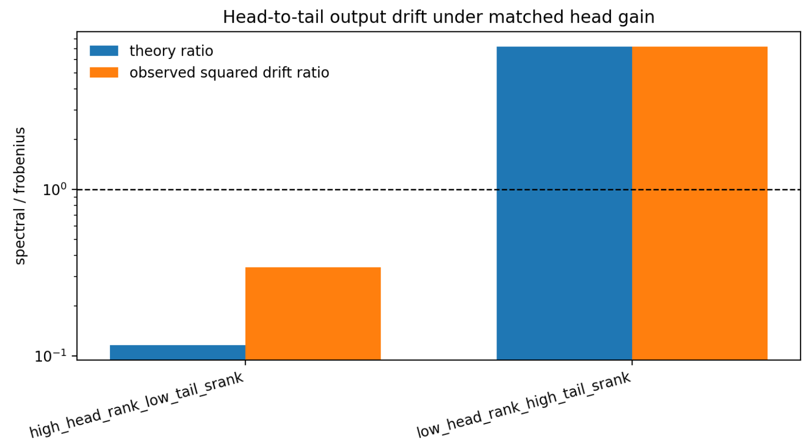

Figure 1: Synthetic head-to-tail boundary diagnostic. The figure compares high-head-rank / low-tail-sensitivity and low-head-rank / high-tail-sensitivity constructions. In this controlled sandwich block, when the head-gradient nuclear rank is above the downstream-aware tail sensitivity rank $\mathrm{ssrank}(B_T, A_T)$, the spectral/polar direction has a lower matched-head-gain squared tail-output drift ratio; reversing the construction reverses the conclusion.

so stronger tail-knowledge claims require checkpoints or benchmarks where the tail function is already useful.

Table 3 also reports the head-alignment and norm readouts behind the matched-gain scaling. In this diagnostic, the spectral/polar head-alignment denominator is 2.083 [1.991, 2.178] times the Frobenius/GD denominator. After scaling to the same head gain, the spectral/Frobenius update has a larger Frobenius norm, 3.112 [2.976, 3.255], but a smaller operator norm, 0.5543 [0.5395, 0.5695]. Thus the mechanism should be described in norm-specific terms rather than as a uniformly smaller parameter step.

Tail loss and margin respond differently from logit drift in this diagnostic. The tail-loss increase difference, spectral minus Frobenius, is 0.001399 [0.0002379, 0.002559], and the tail-margin drop difference is 0.001892 [9.142e-05, 0.003693]. Because the tail-loss interval is positive, lower logit drift does not imply lower cross-entropy loss here; in this diagnostic, spectral/polar slightly increases tail loss more than Frobenius/GD. The absolute scale in Appendix Table 3 shows why this is a caveat rather than a performance result: the tail CE is 11.74 [11.5, 11.99] before the update, and the spectral-minus-Frobenius CE change is only 0.001399 [0.0002379, 0.002559]. This separates the paper's local mechanism from a performance claim: classification loss also depends on the direction of logit movement, including its projection onto the true-class-versus-competing-class margin direction.

5.4 CIFAR-100-LT ResNet18 Architecture Check

To test whether the same one-step readout is specific to the small MLP, we repeat the diagnostic on a CIFAR-100-LT split with classes 0–49 treated as head classes and classes 50–99 as tail classes. The model is a ResNet18 with a CIFAR-style stem. After warmup on the imbalanced train split, the diagnostic fixes BatchNorm state with $\mathrm{eval}()$, forms the head-batch gradient, and applies matched-head-gain Frobenius/GD and spectral/polar interventions only to Conv/Linear matrix weights; convolution kernels are flattened as matrices for the polar direction. This is an architecture robustness check for local drift, not a complete long-tailed benchmark or practical Muon training run.

Across 10 GPU-rerun seeds, spectral/polar has lower squared tail-example logit drift in every paired seed, with spectral/Frobenius ratio 0.5611 [0.5224, 0.6026]. The centered-logit ratio is 0.5628 [0.5242, 0.6041], and the margin-delta ratio is 0.5442 [0.5036, 0.588]. A smaller target-head-gain check at $0.002\,\mathcal{L}_H(\theta)$ gives squared drift ratio 0.7761 [0.7585, 0.7941]. A warmup-checkpoint sweep over 4 ResNet checkpoints gives worst squared drift ratio 0.5611 [0.5224, 0.6026]

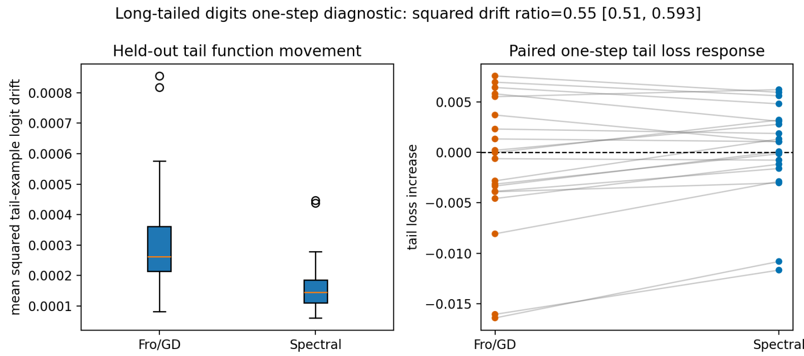

Figure 2: Controlled long-tailed digits one-step diagnostic. The left panel shows that the spectral/polar direction causes lower held-out squared tail-example logit drift after matching head gain. The right panel shows that the tail-loss response does not support an equally strong performance claim.

at 1000 warmup steps. Unlike the small digits one-step diagnostic, the tail-loss increase difference is negative in the default ResNet run, -0.0004206 [-0.0005924, -0.0002487], and remains negative at the smaller target gain, -0.0001719 [-0.0002418, -0.000102], but the tail-accuracy-drop difference $-5e-05$ [-0.0002787, 0.0001787] remains inconclusive. The checkpoint sweep also keeps the claim local: pre-update tail accuracy spans only 0.01915 to 0.068, so these results reduce architecture and checkpoint-selection concerns without proving preservation of a high-quality tail predictor.

We also record a natural-task rank-side proxy on the ResNet checkpoint sweep. Across 40 seed/checkpoint points, the mean matrix-gradient nuclear rank has Pearson correlation 0.7594 [0.6657, 0.8807] and Spearman correlation 0.7366 [0.5493, 0.8383] with the log squared drift ratio. This positive association does not contradict the theorem, because the theorem compares $\mathrm{nrank}(G_H)$ to a downstream-aware tail sensitivity quantity rather than using head-gradient rank alone. The proxy scatter is therefore a boundary check: it shows why natural-task claims require measuring tail-side sensitivity, not only head-gradient rank.

For the final classifier layer, the downstream-aware term can be measured directly from the tail feature matrix feeding $fc.weight$. A final-layer-only ResNet condition diagnostic over 40 seed/checkpoint points gives weakest mean $\mathrm{nrank}(G_H)/\,\mathrm{srank}(H_T)$ score 6.566, condition-favors-spectral fraction 1, and worst final-layer squared drift ratio 0.2391 [0.2201, 0.2597]. This is closer to the theorem than the rank-only proxy, but by itself it is a classifier-layer diagnostic.

Finally, we run a tail-rich checkpoint-quality control with 300 tail-train examples per class. This control is not a standard long-tailed benchmark, but it addresses whether the ResNet drift result only protects a weak tail function. The best pre-update tail accuracy rises to 0.3739 [0.3454, 0.4024], while the worst spectral/Frobenius squared drift ratio remains 0.7292 [0.6923, 0.768]. Appendix Figure 17 adds an explicit tail-count sweep over 4 settings; the worst squared drift ratio is 0.6075 [0.3943, 0.936] at tail count 100, while the best pre-update tail accuracy is 0.3297 [0.2954, 0.3639] at tail count 300. This sweep supports the local drift readout across tail frequencies but has mixed tail-loss signs. At the tail-rich 5000-step checkpoint, an all-layer finite-difference JVP diagnostic probes 21 Conv/Linear weights and 210 paired layer/seed points. Its overall observed squared drift ratio is 0.2011 [0.1845, 0.2192], the scaled-JVP ratio is 0.065 [0.06008, 0.07031], and $21/21$ per-layer observed CI upper endpoints are below one. Thus the lower-drift readout persists when the tail function is more meaningful and when earlier convolutional blocks are probed, but still should not be read as an optimizer-level tail-accuracy result.

5.5 Muon-Style Direction Compatibility

The next diagnostic connects the ideal polar direction to practical Muon-style update directions, while keeping the claim explicitly local. The theory uses the exact polar direction of the current head gradient, $\mathrm{polar}(G_t)$, whereas practical Muon-style updates use a momentum state and a finite Newton–Schulz approximation, closer to $\mathrm{polar}(M_t)$ or $\mathrm{NS}(M_t)$. At a fixed controlled long-

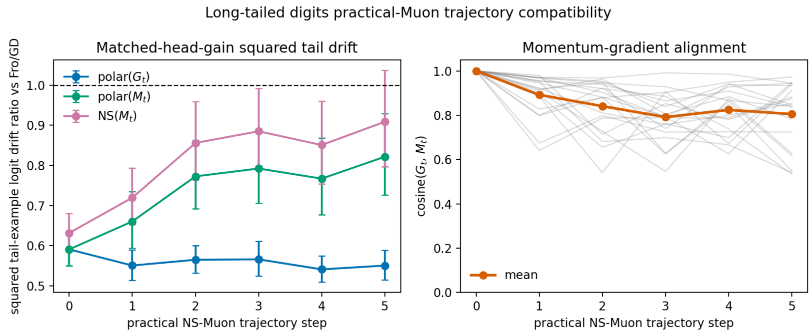

Figure 3: Muon-style direction compatibility diagnostic along a short NS-Muon-style trajectory. The left panel reports the matched-head-gain squared tail-example logit drift ratio at each sampled trajectory state. The right panel shows momentum-gradient alignment over trajectory steps. The figure evaluates local compatibility of Muon-style directions with the spectral/polar mechanism and does not measure final tail accuracy or full-optimizer benchmark performance.

tailed digits checkpoint, momentum polar has a matched-head-gain squared drift ratio below 1: the $\mathrm{polar}(M_t)$ squared drift ratio is 0.8199 [0.6951, 0.9672]. The finite Newton–Schulz direction is weaker, with squared drift ratio 0.9116 [0.7696, 1.08]; because this interval crosses 1, this particular diagnostic is not stable evidence that the finite-step NS direction alone lowers drift.

We then run a 6-step NS-Muon-style head-only trajectory and re-run the matched-head-gain diagnostic at each sampled state. Across 120 state-step comparisons, the $\mathrm{polar}(M_t)$ squared drift ratio is 0.7292 [0.6891, 0.7717], while the $\mathrm{NS}(M_t)$ squared drift ratio is 0.8019 [0.7583, 0.848]. The mean momentum-gradient cosine is 0.8589 [0.8328, 0.885], so M_t is not identical to G_t. To reduce state-selection concern, Appendix Figure 6 repeats the same diagnostic on states generated by a Fro/GD-style trajectory with the same seeds, batches, length, and nominal trajectory learning rate. On those Fro/GD-generated states, $\mathrm{polar}(M_t)$ gives squared drift ratio 0.7255 [0.686, 0.7672], and $\mathrm{NS}(M_t)$ gives 0.7973 [0.7546, 0.8425].

Appendix Figure 21 repeats this bridge at CIFAR-100-LT ResNet18 trajectory states. Starting from the same tail-rich 5000-step ResNet checkpoint, it samples matrix-weight-only AdamW and finite-step NS-Muon trajectories and re-runs the matched-head-gain direction diagnostic. On AdamW-sampled states, $\mathrm{NS}(M_t)$ has squared drift ratio 0.8628 [0.8154, 0.913]; on NS-Muon-sampled states, it has squared drift ratio 0.7247 [0.676, 0.777]. This strengthens the practical-direction bridge beyond the digits setting, but it remains a local trajectory-state compatibility check, not an optimizer benchmark; the practical-training diagnostic and its learning-rate caveat are moved to Appendix K.

Appendix Figure 20 then adds an actual CIFAR-100-LT ResNet18 final-training pilot for finite Newton–Schulz Muon-style matrix-weight updates. The result is negative under the tested recipes: augmented AdamW reaches all/few balanced accuracy 0.3613 [0.3595, 0.363] and 0.08856 [0.07566, 0.1014], while NS-Muon with learning rate 10^{-4} reaches 0.1265 [0.1218, 0.1312] and 0.0008889 [-0.0006533, 0.002431]. Its paired all/few differences versus augmented AdamW are -0.2348 [-0.2395, -0.23] and -0.08767 [-0.09948, -0.07585]. This does not contradict the local bridge; it shows that the tested final-training NS-Muon recipe is not a competitive optimizer result.

5.6 Layerwise Diagnostic

The layerwise diagnostic separates unit-direction tail sensitivity from matched-gain step-size scaling. For each matrix block W_ℓ in the two-layer MLP, we record $\mathrm{nrank}(G_{H,\ell})$, estimate the unit-direction tail JVP by finite differences, and measure layer-only drift after matching first-order head gain. For the final linear layer, the local map is an exact sandwich block with $\mathrm{nrank}(G_{H,2})/\,\mathrm{ssrank}(B_{T,2}, A_{T,2}) = 2.207$. For the first ReLU layer, gates vary across tail samples, so we report the frozen-gate local-operator score $\mathrm{nrank}(G_{H,1})/\,\mathrm{srank}(J_{T,1}) = 0.7261$.

The unit-JVP squared drift ratio is above 1 in both layers: 1.45 and 1.476, so the spectral/polar unit direction is not inherently less tail-sensitive. After matched-head-gain scaling, however, the scaled-JVP and observed squared drift ratios agree: 0.4766 and 0.4767 in layer 1, and 0.596 and 0.596 in layer 2. Table 4 gives intervals. The mechanism is therefore more precise than "spectral directions are always safer": in this experiment, the lower observed drift comes from smaller matched-gain step scaling.

The corresponding layerwise figure is reported in Appendix K.

6 Limitations and Discussion

6.1 What the Evidence Establishes

Table 2 separates the current diagnostic claims from stronger claims that require additional experiments. The claim boundary is local: after matching head-batch gain, spectral/polar directions can reduce squared logit drift on held-out tail examples in the tested settings. The Muon results are selected-state compatibility checks from ideal polar directions to Muon-style directions, not evidence that Muon is generally better on long-tailed classification.

The detailed claim boundary is reported in Appendix Table 2. The main limitations are: lower squared logit drift on held-out tail examples does not imply lower tail cross-entropy or higher accuracy; fixed-checkpoint Muon-style directions are local compatibility checks rather than a theory of complete Muon training; the synthetic boundary validates a mechanism sign, not a natural-task predictor; and layerwise evidence shows that the lower observed drift comes from matched-head-gain scaling, not from uniformly lower unit-direction tail sensitivity.

6.2 Claim Scope and Rebuttal Discipline

The supported claim is local matched-head-gain tail-example logit drift: after equalizing first-order progress on a head batch, the tested spectral/polar directions can perturb held-out tail-example logits less than Frobenius/GD in the reported diagnostics. The theorem is a local worst-case comparison for a fixed linearization and a specified tail-sensitivity operator, not a global optimizer theorem, a convergence result, or a final-accuracy guarantee. Consequently, any claim about tail loss, margin, accuracy, or practical Muon training must be read as a separate empirical question.

We also separate supported claims from registered-but-not-yet-supportive tests. The current v5 condition-score program is a registered pending test: the score family is frozen, but the unspent final architecture and data splits must pass before the paper can claim a successful natural-task predictor of layer risk. The natural negative-search readout is complete only for its registered phase1 family: the interim audit has observed $26/26$ settings, `raw_worse_rows=0`, and no adjusted primary full-drift failure, so it is a finite registered phase1 null candidate with detectable-effect and tail-quality caveats. This does not prove that no natural counterexample exists outside the registered phase1 space. The practical CIFAR-100-LT NS-Muon final-training pilot is a negative benchmark boundary, not evidence for an optimizer-performance advantage. Finally, the server validation and rendered-PDF fallback are current reproduction evidence, while preferred LaTeX clean-checkout reproduction remains an explicit gate.

Additional discussion of the gap between ideal spectral gradients and practical Muon-style updates, the role of mini-batch noise, and the remaining experiments needed for optimizer-level claims is deferred to Appendix H.

7 Conclusion

This paper develops a function-drift view of long-tailed small-batch training: match head progress, then measure tail drift. The core technical object is the head-to-tail interference coefficient, which compares update geometries by their worst-case tail-output drift per unit first-order head gain. For a sandwich matrix block $J_T(D) = B_T D A_T$, this comparison yields the condition $\mathrm{nrank}(G_H) > \mathrm{ssrank}(B_T, A_T)$: spectral geometry is favored when the head gradient is sufficiently spectrally spread relative to downstream tail sensitivity.

The experiments support this local mechanism while also delimiting it. The controlled synthetic boundary reverses the drift ordering when the rank condition is reversed, changing the spectral/Frobenius squared drift ratio from 0.3403 to 7.208. The controlled long-tailed digits diagnostics then show lower matched-gain squared tail-example logit drift for spectral/polar and selected Muon-style directions, with one-step squared drift ratio 0.5501 [0.5101, 0.5931] and short-trajectory $\mathrm{NS}(M_t)$ squared drift ratio 0.8019 [0.7583, 0.848]. The CIFAR-100-LT ResNet18 architecture check gives squared drift ratio 0.5611 [0.5224, 0.6026], the final-layer condition diagnostic gives worst classifier-layer drift ratio 0.2391 [0.2201, 0.2597], the tail-rich control keeps worst drift ratio 0.7292 [0.6923, 0.768] while raising tail accuracy to 0.3739 [0.3454, 0.4024], the tail-count imbalance sweep has worst drift ratio 0.6075 [0.3943, 0.936], the all-layer JVP control gives observed drift ratio 0.2011 [0.1845, 0.2192], and the ResNet practical bridge keeps finite-step $\mathrm{NS}(M_t)$ below Fro/GD on both AdamW- and NS-Muon-sampled states. An actual CIFAR-100-LT ResNet18 NS-Muon final-training pilot is negative: the tested lr 10^{-4} recipe has all-group balanced-accuracy difference -0.2348 [-0.2395, -0.23] versus augmented AdamW. These results make the mechanism measurable, but not a performance theorem: lower logit drift need not imply lower tail cross-entropy, larger margin, or higher accuracy, and practical Muon introduces momentum, approximation, schedule, and tuning effects. The paper is therefore a theory and initial empirical study of head-to-tail function drift, not a complete long-tailed classification optimizer benchmark.

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The appendix provides supporting definitions and checks in the order they are used in the paper: notation and norm preliminaries; proofs and the matrix-block derivation; multi-step drift and margin-certificate consequences; the relation to existing layerwise spectral-update theory; blockwise spectral geometry for the MLP diagnostics; additional discussion of practical Muon-style directions; reproducibility details; auxiliary tables; and auxiliary figures.

A PRELIMINARIES

A.1 NOTATION SUMMARY

Table 1: Notation summary for the head-to-tail diagnostic.

Symbol	Meaning
H, T	The head batch that defines the update and the tail probe batch whose outputs are monitored.
g_H	The full-parameter gradient of the head-batch loss, $g_H = \nabla_\theta \mathcal{L}_H(\theta)$.
G_H	The matrix-block gradient of the head-batch loss, $G_H = \nabla_W \mathcal{L}_H$.
D	A matrix-block perturbation, $D = \Delta W$.
J_T	The local tail-output Jacobian or linear tail-output perturbation operator.
A_T	The tail activation entering the perturbed matrix block; it is a data-dependent local activation matrix, not optimizer state.
B_T	The downstream tail Jacobian after the perturbed block; together with A_T, it defines the sandwich map $J_T(D) = B_T D A_T$.
ρ	The matched first-order head gain imposed before comparing tail drift.
$\mathrm{nrank}(G_H)$	The nuclear rank of the head matrix gradient, $\|G_H\|_*^2 \,/\, \|G_H\|_F^2$.
$\mathrm{srank}(A_T)$	The stable rank of the tail activation matrix, $\|A_T\|_F^2 \,/\, \|A_T\|_{\mathrm{op}}^2$.
$\mathrm{ssrank}(B_T, A_T)$	The downstream-aware stable-rank quantity formed by pairing the singular spectra of B_T and A_T: $\sum_i \sigma_i(B_T)^2 \sigma_i(A_T)^2 / [\sigma_1(B_T)^2 \sigma_1(A_T)^2]$.

A.2 MATRIX NORMS

For a matrix M, write its singular values as

$$\sigma_1(M) \geq \sigma_2(M) \geq \cdots .$$

The spectral norm is

$$\|M\|_{\mathrm{op}} = \sigma_1(M),$$

the Frobenius norm is

$$\|M\|_F = \left(\sum_i \sigma_i(M)^2 \right)^{1/2},$$

and the nuclear norm is

$$\|M\|_* = \sum_i \sigma_i(M).$$

The stable rank of a matrix is

$$\mathrm{srank}(M) = \frac{\|M\|_F^2}{\|M\|_{\mathrm{op}}^2}.$$

The nuclear rank of a gradient is

$$\mathrm{nrank}(G) = \frac{\|G\|_*^2}{\|G\|_F^2}.$$

A.3 SPECTRAL GRADIENT DIRECTIONS

Let a matrix gradient be

$$G = U \Sigma V^\top .$$

The Frobenius-normalized gradient direction is

$$u_F = \frac{G}{\|G\|_F}.$$

The spectral-gradient, or polar, direction is

$$u_S = UV^\top.$$

By duality between the spectral norm and the nuclear norm,

$$\max_{\|D\|_{\text{op}} \le 1} \langle G, D \rangle = \|G\|_* ,$$

with optimizer

$$D = UV^\top.$$

By self-duality of the Frobenius norm,

$$\max_{\|D\|_F \le 1} \langle G, D \rangle = \|G\|_F ,$$

with optimizer

$$D = \frac{G}{\|G\|_F}.$$

Thus the polar update is the first-order steepest direction under spectral-norm geometry, while the normalized gradient is the first-order steepest direction under Frobenius geometry.

B Proof Details

Proof of the matched-gain drift bound. Let u_N be a norm-N steepest direction:

$$\|u_N\|_N = 1, \qquad \langle g_H, u_N \rangle = \|g_H\|_{N,*} .$$

Set

$$\Delta_N = -\frac{\rho}{\|g_H\|_{N,*}} u_N.$$

Then $-\langle g_H, \Delta_N \rangle = \rho$. By the definition of $K_{T,N}$,

$$\|J_T \Delta_N\| = \frac{\rho}{\|g_H\|_{N,*}} \|J_T u_N\| \le \frac{\rho K_{T,N}}{\|g_H\|_{N,*}}.$$

Squaring proves

$$\|J_T \Delta_N\|^2 \le \rho^2 \mathcal{I}_N(T \mid H).$$

□

Lemma B.1 (Sandwiched sensitivity over the spectral-norm unit ball)**.** *For any matrices B, A,*

$$\sup_{\|D\|_{\text{op}} \le 1} \|BDA\|_F^2 = \sum_i \sigma_i(B)^2 \sigma_i(A)^2,$$

where singular values are sorted in descending order and padded with zeros after the shorter sequence.

Proof. Let

$$B^\top B = P \operatorname{diag}(\beta) P^\top, \qquad AA^\top = Q \operatorname{diag}(\alpha) Q^\top,$$

where $\beta_i = \sigma_i(B)^2$ and $\alpha_i = \sigma_i(A)^2$. For $\|D\|_{\text{op}} \le 1$, define $C = P^\top D Q$. Then $\|C\|_{\text{op}} \le 1$, and

$$\|BDA\|_F^2 = \sum_{i,j} \beta_i \alpha_j C_{ij}^2.$$

The matrix $M_{ij} = C_{ij}^2$ is doubly substochastic because $CC^\top \preceq I$ and $C^\top C \preceq I$. Maximizing the linear objective $\sum_{i,j} \beta_i \alpha_j M_{ij}$ over this polytope is achieved by a partial permutation, and the rearrangement inequality pairs the largest β_i with the largest α_i. This gives the upper bound $\sum_i \beta_i \alpha_i$. The bound is attained by choosing C to be a partial isometry between the corresponding top eigendirections and setting $D = PCQ^\top$. □

C Matrix-Block Derivation

This appendix expands the matrix-block calculation summarized in the main text. Let W be a matrix parameter block and let $G_H = \nabla_W \mathcal{L}_H$ be the head-batch matrix gradient. Assume the tail perturbation of this block has the sandwich form

$$J_T(D) = B_T D A_T,$$

where A_T is the activation matrix entering the block on tail data, B_T is the downstream tail Jacobian, and $D = \Delta W$. If $D \in \mathbb{R}^{m\times k}$, we use the convention

$$B_T \in \mathbb{R}^{q\times m}, \qquad A_T \in \mathbb{R}^{k\times r}, \qquad B_T D A_T \in \mathbb{R}^{q\times r}.$$

For a general nonlinear block, the local map can instead be written as $\operatorname{vec}(\mathcal{J}_{T,\ell}(D)) = L_{T,\ell}\operatorname{vec}(D)$; the Kronecker form $L_{T,\ell} = A_T^\top \otimes B_T$ is the sandwich special case.

Under Frobenius geometry, the steepest direction is $u_F = G_H / \|G_H\|_F$, the dual norm is $\|G_H\|_F$, and

$$\operatorname{vec}(B_T D A_T) = (A_T^\top \otimes B_T)\operatorname{vec}(D).$$

Therefore the Frobenius unit-step tail sensitivity is

$$K_{T,F}^2 = \sup_{\|D\|_F \le 1} \|B_T D A_T\|_F^2 = \|B_T\|_{\mathrm{op}}^2 \|A_T\|_{\mathrm{op}}^2,$$

which gives

$$\mathcal{I}_F(T \mid H) = \frac{\|B_T\|_{\mathrm{op}}^2 \|A_T\|_{\mathrm{op}}^2}{\|G_H\|_F^2}.$$

Under spectral geometry, write $G_H = U\Sigma V^\top$. The spectral-gradient direction is $u_S = UV^\top$, and the dual norm is $\|G_H\|_*$. By Lemma B.1, the spectral-norm unit-ball sensitivity is

$$K_{T,S}^2 = \sup_{\|D\|_{\mathrm{op}} \le 1} \|B_T D A_T\|_F^2 = \sum_i \sigma_i(B_T)^2 \sigma_i(A_T)^2,$$

where singular values are sorted in descending order and padded with zeros. Hence

$$\mathcal{I}_S(T \mid H) = \frac{\sum_i \sigma_i(B_T)^2 \sigma_i(A_T)^2}{\|G_H\|_*^2}.$$

Spectral geometry has the smaller worst-case matched-gain tail-drift bound when $\mathcal{I}_S(T \mid H) < \mathcal{I}_F(T \mid H)$. Substituting the expressions above and rearranging gives

$$\frac{\|G_H\|_*^2}{\|G_H\|_F^2} > \frac{\sum_i \sigma_i(B_T)^2 \sigma_i(A_T)^2}{\sigma_1(B_T)^2 \sigma_1(A_T)^2}.$$

We call the right-hand side the downstream-aware stable-rank quantity:

$$\operatorname{ssrank}(B_T, A_T) = \frac{\sum_i \sigma_i(B_T)^2 \sigma_i(A_T)^2}{\sigma_1(B_T)^2 \sigma_1(A_T)^2}.$$

This bound ratio is defined for nondegenerate tail-sensitive sandwich blocks with $\sigma_1(B_T)\sigma_1(A_T) > 0$. If $\sigma_1(B_T)\sigma_1(A_T) = 0$, then $B_T D A_T = 0$ for all D, the sandwich tail sensitivity is zero, and the rank ratio is not an informative boundary. Since $\operatorname{nrank}(G_H) = \|G_H\|_*^2 / \|G_H\|_F^2$, the condition becomes

$$\operatorname{nrank}(G_H) > \operatorname{ssrank}(B_T, A_T).$$

D Tail Drift across Multiple Head-Only Steps

In long-tailed small-batch training, tail batches are often separated by several head-only steps. Suppose a tail batch appears every n head batches on average. During these n head-only steps, let the parameter updates be

$$\Delta_1, \Delta_2, \ldots, \Delta_n.$$

The accumulated tail-output drift is

$$F_T(\theta_n) - F_T(\theta_0) \approx \sum_{t=1}^{n} J_T(\theta_t)\Delta_t.$$

If the local approximation

$$J_T(\theta_t) \approx J_T$$

holds, then

$$F_T(\theta_n) - F_T(\theta_0) \approx J_T \sum_{t=1}^{n} \Delta_t.$$

In the worst case,

$$\|F_T(\theta_n) - F_T(\theta_0)\| \le \sum_{t=1}^{n} \|J_T\Delta_t\|.$$

If each step obtains the same first-order head gain ρ and uses the norm-N steepest update, then

$$\|J_T\Delta_t\| \le \rho\sqrt{\mathcal{I}_N(T \mid H_t)}.$$

so

$$\boxed{\|F_T(\theta_n) - F_T(\theta_0)\| \le \rho \sum_{t=1}^{n} \sqrt{\mathcal{I}_N(T \mid H_t)}.}$$

If we further approximate

$$\mathcal{I}_N(T \mid H_t) \approx \mathcal{I}_N,$$

then

$$\boxed{\|F_T(\theta_n) - F_T(\theta_0)\| \lesssim n\rho\sqrt{\mathcal{I}_N}.}$$

This shows why tail rarity matters for the local mechanism: the larger n is, the more important it is for the update geometry to have small head-to-tail interference. Otherwise, the model may undergo substantial tail drift before tail samples appear again.

E Tail Margin Certificate

Controlling tail-output drift alone is not enough; we also want tail predictions to remain unchanged. Consider a tail sample x with true class y and logits $f(x)$. Define the tail margin:

$$m_T(x) = f_y(x) - \max_{k \ne y} f_k(x).$$

If a logit perturbation $\delta f(x)$ satisfies

$$\|\delta f(x)\|_\infty < \frac{1}{2} m_T(x),$$

then the classification of this sample is unchanged. This is a margin certificate only for samples that are currently classified correctly with $m_T(x) > 0$. If $m_T(x) \le 0$, the sample is already non-positive-margin and this bound does not certify unchanged prediction.

Let the minimum margin on the tail batch be

$$m_T^{\min} = \min_{x \in T} m_T(x).$$

If

$$\|F_T(\theta_n) - F_T(\theta_0)\|_\infty < \frac{1}{2} m_T^{\min},$$

then all predictions on the tail batch remain unchanged.

Using the drift bound from the previous section, if

$$n\rho\sqrt{\mathcal{I}_N^\infty} < \frac{1}{2} m_T^{\min},$$

then head-only updates do not change the predictions on the tail batch. Here $\mathcal{I}_N^\infty$ is the head-to-tail interference coefficient defined using logit ℓ_∞ drift.

This gives a sufficient unchanged-prediction condition on the tail batch:

$$\boxed{n\rho\sqrt{\mathcal{I}_N^\infty} < \frac{1}{2} m_T^{\min}.}$$

It says that the rarer the tail data are, or the larger each head step is, the more important it is for the matched-gain update rule to have small head-to-tail interference.

F Relation to Existing Layerwise Spectral-Update Theory

Existing spectral-gradient theory usually studies the advantage of spectral updates on the same batch. For a linear block $Y = WA$, with current gradient G, the one-step decreases of Frobenius and spectral updates are approximately

$$\mathrm{Dec}_F \asymp \frac{\|G\|_F^2}{\|A\|_{\mathrm{op}}^2},$$

and

$$\mathrm{Dec}_S \asymp \frac{\|G\|_*^2}{\|A\|_F^2}.$$

Thus, the same-batch spectral-decrease bound is larger than the Frobenius bound when

$$\mathrm{nrank}(G) > \mathrm{srank}(A).$$

This paper studies a different question. We do not ask whether the spectral update decreases the loss more on the same batch. Instead, we ask:

after matching head-batch gain, which update perturbs the tail function less?

Therefore our condition is not

$$\mathrm{nrank}(G) > \mathrm{srank}(A_H),$$

but

$$\boxed{\mathrm{nrank}(G_H) > \mathrm{ssrank}(B_T, A_T).}$$

The left-hand side comes from the head gradient, while the right-hand side comes from tail functional sensitivity. This is a head-to-tail cross-distribution condition.

G Blockwise Spectral Geometry

The MLP diagnostics use blockwise polar directions, one matrix direction per layer. This direction has a simple norm interpretation. For matrix blocks $D = (D_1, \ldots, D_L)$, define

$$\|D\|_{\mathrm{maxop}} = \max_\ell \|D_\ell\|_{\mathrm{op}}.$$

Proposition G.1 (Blockwise polar steepest direction)**.** *Let $G = (G_1, \ldots, G_L)$ be a collection of matrix gradients and let $\|D\|_{\mathrm{maxop}} = \max_\ell \|D_\ell\|_{\mathrm{op}}$. The dual norm is*

$$\|G\|_{*,1} = \sum_\ell \|G_\ell\|_*,$$

and the blockwise polar direction $D_\ell = \mathrm{polar}(G_\ell)$ for each nonzero block is a maximizer of the linearized gain over the $\|\cdot\|_{\mathrm{maxop}}$ unit ball:

$$\sup_{\|D\|_{\mathrm{maxop}} \le 1} \sum_\ell \langle G_\ell, D_\ell \rangle = \sum_\ell \|G_\ell\|_*.$$

Proof. For any feasible D, von Neumann's trace inequality gives $\langle G_\ell, D_\ell \rangle \leq \|G_\ell\|_* \|D_\ell\|_{\mathrm{op}} \leq \|G_\ell\|_*$ for each block. Summing over blocks gives the upper bound $\sum_\ell \|G_\ell\|_*$. If $G_\ell = U_\ell \Sigma_\ell V_\ell^\top$, choosing $D_\ell = U_\ell V_\ell^\top = \mathrm{polar}(G_\ell)$ for each nonzero block has $\|D_\ell\|_{\mathrm{op}} = 1$ and attains equality in every block, so it attains the supremum. □

Thus the empirical direction $d_S = \{\mathrm{polar}(G_\ell)\}_\ell$ is the steepest direction for this blockwise spectral geometry. In the neural diagnostics we still scale the resulting full direction by the actual head alignment $\langle g_H, d_S \rangle$, so the reported comparisons remain matched-head-gain measurements rather than a claim about complete optimizer dynamics.

H Additional Discussion

The core theory applies to the idealized spectral gradient direction $\mathrm{polar}(G) = UV^\top$. Practical Muon uses momentum and a finite Newton–Schulz approximation, so its matrix direction is closer to $\mathrm{polar}(M_t)$ or $\mathrm{NS}(M_t)$ than to $\mathrm{polar}(G_t)$. A faithful Muon analysis must therefore include a momentum-gradient alignment factor and a state-distribution question: which checkpoints are being probed? Figures 3 and 5 show selected-state compatibility checks, and Figure 6 repeats the short-trajectory diagnostic on Fro/GD-generated states to reduce state-selection concern. Figure 11 is a controlled practical-training diagnostic kept in Appendix K. These observations motivate using Muon-style directions as local implementation probes for spectral/polar geometry, while leaving a full theory of Muon training to future work.

In the practical diagnostic, the Muon-style run has a final train-loss ratio of 0.6468 [0.604, 0.6926] and a tail-margin difference of 1.955 [1.628, 2.281] relative to Adam, but this evidence is insufficient for an optimizer-level comparison. The learning-rate sweep limits the interpretation: at the largest tested Muon learning rate, the tail-loss ratio rises to 2.049. We therefore treat this experiment as a consistency check for the local mechanism, not as a tuned Adam-vs-Muon comparison.

A full polar update equalizes all nonzero singular directions. If small singular directions of the head gradient are mini-batch noise, spectral geometry may increase tail drift. This is one reason the paper focuses on matched-head-gain diagnostics rather than claiming a universal optimizer advantage. In long-tailed learning, the proposed mechanism is tail-function drift reduction between rare tail batches. Whether this becomes lower tail loss, larger margin, or higher tail accuracy requires longer training studies and standard long-tailed benchmarks.

The current evidence supports a local mechanism claim within the tested matched-head-gain diagnostics, not a complete optimizer benchmark. The CIFAR-100-LT ResNet18 result is a stronger architecture check than the small MLP diagnostics, and the tail-rich all-layer JVP control reduces the concern that only the classifier layer was probed. Moving from mechanism to optimizer-level claims still requires longer tuned runs on standard long-tailed benchmarks, and practical Muon training needs more complete ablation over schedules, weight decay, and checkpoints. For future work, larger-architecture layerwise diagnostics should turn the current finite-difference JVP bridge into a pre-registered predictive condition test across held-out checkpoints, architectures, and datasets. The important next test is not only whether spectral geometry lowers matched-gain drift, but when the measurable condition accounts for tail drift and when it fails because of mini-batch noise, momentum misalignment, or layerwise sensitivity structure.

I Reproducibility Details

This appendix specifies the reproduction parameters used for the main experiments. The synthetic head-to-tail boundary experiment uses 80 random seeds and explicitly controls the singular values and aligned singular-vector structure of G_H, A_T, B_T. Because the construction fixes the relevant spectra and their alignment, the reported ratios are effectively deterministic across seeds and the intervals collapse to point intervals. The controlled long-tailed digits matched-head-gain experiments use 20 seeds and the scikit-learn 8×8 digits dataset (Pedregosa et al., 2011). Across all default seeds, digits classes 0–4 are fixed as head classes and classes 5–9 are fixed as tail classes; the random seed only changes the within-class sample split, mini-batches, noise draws, and initialization. The training split uses 100 samples per head class and 40 samples per tail class, and the held-out tail evaluation set uses 40 samples per tail class. This is a controlled head-heavy digits diagnostic rather than a standard long-tailed benchmark. The model is a two-layer MLP with hidden dimension 32 and no bias, batch-normalization, classifier-only head, or other non-matrix parameters.

The long-tailed digits checkpoint is obtained by 80 noisy mini-batch warmup steps, with batch size 64 and input-noise standard deviation 0.05. The fixed-checkpoint one-step, layerwise, and fixed-checkpoint Muon-style diagnostics then form the head gradient on a clean head mini-batch and measure drift, activations, ranks, and losses on the clean held-out tail probe set. The short practical NS-Muon trajectory forms each trajectory head gradient on a noisy head mini-batch before re-running the matched-gain diagnostic at that state. The Muon state-source control repeats this trajectory-level diagnostic on Fro/GD-style trajectory states, using the same seeds, batch draws, trajectory length, and nominal trajectory learning rate, to check whether the Muon-style direction readout is unique to NS-Muon-generated states. The imbalance ablation repeats the one-step diagnostic with tail training examples per class $80, 40, 20, 10$, while keeping the held-out tail evaluation set at 40 examples per tail class. The checkpoint sweep repeats the one-step diagnostic after $20, 40, 80, 120, 160$ warmup steps. The class-partition sweep repeats it across five balanced 5-class head / 5-class tail partitions. The one-step and layerwise matched-head-gain experiments use

$$\rho = 0.02\,\mathcal{L}_H(\theta)$$

as the target first-order head gain. Let G_ℓ denote the head-batch gradient for matrix layer ℓ. The Frobenius/GD direction is $d_F = \{G_\ell\}_\ell / \sqrt{\sum_\ell \|G_\ell\|_F^2}$, while the spectral/polar direction is $d_S = \{\mathrm{polar}(G_\ell)\}_\ell$. For a Muon-style direction, G_ℓ is replaced by the momentum state M_ℓ, and the finite Newton–Schulz direction is computed blockwise after reshaping each parameter tensor to its matrix view. All directions are scaled by $\Delta_d = -\rho d / \langle g_H, d\rangle$, using the actual head-gradient alignment of the full direction.

The CIFAR-100-LT ResNet18 one-step diagnostic uses 10 seeds, head classes 0–49, tail classes 50–99, 300 head-train examples per class, 30 tail-train examples per class, and 40 held-out tail-evaluation examples per class from the CIFAR-100 test split. The default checkpoint is warmed up for 1000 AdamW steps with batch size 256, learning rate 3×10^{-4}, and weight decay 10^{-4}, then evaluated on GPU in float32 with target head gain $0.005\,\mathcal{L}_H(\theta)$, plus a smaller-head-gain check at $0.002\,\mathcal{L}_H(\theta)$. The checkpoint sweep repeats the $0.005\,\mathcal{L}_H(\theta)$ diagnostic at 250, 500, 1000, and 2000 warmup steps. The ResNet18 uses a CIFAR-style 3×3 stride-1 stem and no max-pool. During the one-step diagnostic, BatchNorm state is fixed with $\mathrm{eval}()$, and only Conv/Linear matrix weights are intervened on; convolution kernels are flattened to output-channel by input-channel-times-kernel-size matrices for the polar direction. The condition-proxy scatter reuses the checkpoint-sweep outputs and reports correlations between seed/checkpoint drift ratios and mean Conv/Linear matrix-gradient nuclear rank; it does not estimate the downstream-aware $\mathrm{ssrank}(B_T, A_T)$ term. The final-layer condition scatter uses the same warmup schedule and $\rho = 0.005\,\mathcal{L}_H(\theta)$, freezes all earlier layers, updates only $fc.weight$, and measures $\mathrm{nrank}(G_H)/\,\mathrm{srank}(H_T)$ from the tail feature matrix H_T. The tail-quality control changes only the train split and warmup schedule: it uses 300 tail-train examples per class and warmup checkpoints 2000, 5000, and 10000, then applies the same matched-head-gain diagnostic. The imbalance sweep keeps 300 head-train examples per class, uses tail counts 10, 30, 100, and 300 with 4 settings and 3 seeds, holds tail evaluation at 40 examples per class, warms up for 1000 AdamW steps, and applies the same $0.005\,\mathcal{L}_H(\theta)$ matched-head-gain diagnostic. The all-layer JVP diagnostic uses the same tail-rich split at 5000 warmup steps, probes all Conv/Linear weights one layer at a time, estimates unit-direction tail JVP by finite difference with $\epsilon = 10^{-4}$, and measures both scaled-JVP and observed layer-only drift after matching the same first-order head gain. The checkpoint-transfer version repeats the all-layer JVP probe at 2000, 5000, and 10000 warmup steps and asks whether source-checkpoint layer scores predict held-out checkpoint observed layer drift; its residual test fits source observed log drift from log early-layer prior and applies that source fit to the target checkpoint before ranking residual risk. The ResNet practical Muon bridge starts from the same 5000-step tail-rich checkpoint, samples three matrix-weight-only trajectory states from AdamW and finite-step NS-Muon updates, and at each state compares Fro/GD, $\mathrm{polar}(G_t)$, $\mathrm{polar}(M_t)$, and $\mathrm{NS}(M_t)$ after matching the same head-batch first-order gain. The ResNet NS-Muon final-training pilot uses an IF=100 CIFAR-100-LT split, random crop/flip augmentation, batch size 256, 5000 training steps, 3 seeds, augmented AdamW as the baseline, finite Newton–Schulz steps on Conv/Linear matrix weights, and AdamW on non-matrix parameters. It is evaluated by final many/medium/few/all balanced accuracy and paired differences versus augmented AdamW.

In the head-only forgetting experiment, the head mini-batches and target gains are precomputed once per seed from the initial checkpoint before either geometry is rolled out. Specifically, for each scheduled head-only step t, the experiment stores the same noisy head batch and the same reference target $\rho_t = 0.02\,\mathcal{L}_H(\theta_0; B_t)$ for both Frobenius/GD and spectral/polar trajectories. The two trajectories then start from the same checkpoint and apply their own matched-gain directions using that shared ρ_t sequence. Thus the intended first-order head-gain budget is matched per seed and step; the realized nonlinear head-loss decrease is measured after each update and is not assumed to be exactly identical. Layerwise finite-difference JVP uses $\epsilon = 10^{-4}$.

The finite Newton–Schulz approximation normalizes each matrix by its spectral norm, returns the zero direction for a zero matrix, and otherwise iterates

$$X_{k+1} = 1.5X_k - 0.5X_k X_k^\top X_k$$

for the configured number of steps. The practical imbalanced-training diagnostic does not use the matched-head-gain protocol. Instead, it directly compares Adam and finite-Newton–Schulz Muon-style updates from the same initialization under the same noisy mini-batch schedule. This experiment uses 10 training samples per tail class and keeps 40 held-out tail-evaluation samples per tail class. Training runs for 80 noisy mini-batch steps. Adam uses learning rate 0.01, and the Muon-style update uses learning rate 0.03, momentum $\beta = 0.9$, 5 Newton–Schulz approximation steps, no weight decay, no learning-rate schedule, and no gradient clipping.

All confidence intervals reported in this paper are across-seed paired estimates. For ratio metrics, we first compute the spectral/Frobenius or Muon/Adam ratio within each seed, compute a 95% confidence interval on the log scale, and map the result back to the original scale. For trajectory diagnostics with multiple sampled states per seed, we first average log-ratios over sampled states within each seed, then compute the paired CI across seeds. For difference metrics, such as tail-loss increase difference or margin difference, we compute a 95% confidence interval directly on the paired difference, with the same seed-level aggregation for trajectory diagnostics.

J Additional Empirical Tables

Table 2: Claim boundary. All reported numbers come from matched-head-gain diagnostics unless explicitly marked as practical training; intervals are paired confidence intervals.

Claim	Evidence	Boundary
Matched-head-gain spectral/polar directions can reduce squared logit drift on held-out tail examples.	Controlled long-tailed digits one-step squared drift ratio 0.5501 [0.5101, 0.5931]; spectral/Fro head-alignment ratio 2.083 [1.991, 2.178]; matched operator-norm ratio 0.5543 [0.5395, 0.5695]; 8-step final squared drift ratio 0.6167 [0.5744, 0.6622].	Does not imply tail loss, margin, or accuracy must improve; the one-step CE tail-loss interval is positive, and matched scaling is norm-specific rather than a uniformly smaller parameter step.
$\mathrm{polar}(M_t)$ and $\mathrm{NS}(M_t)$ provide selected-state compatibility checks for Muon-style directions.	Fixed checkpoint: $\mathrm{polar}(M_t)$ squared drift ratio 0.8199 [0.6951, 0.9672], NS squared drift ratio 0.9116 [0.7696, 1.08]; short trajectory: NS squared drift ratio 0.8019 [0.7583, 0.848]; Fro/GD-state control: NS squared drift ratio 0.7973 [0.7546, 0.8425].	Local compatibility only; fixed-checkpoint NS has a CI crossing 1, and the Fro/GD-state control reduces but does not eliminate state-selection risk.
$\mathrm{nrank}(G_H) > \mathrm{ssrank}(B_T, A_T)$ is a discriminative mechanism boundary.	Synthetic positive-boundary squared drift ratio 0.3403 [0.3403, 0.3403]; negative-boundary squared drift ratio 7.208 [7.208, 7.208], with direction matching the condition.	Mechanism boundary check; does not predict standard benchmark performance.
Layerwise drift reduction mainly comes from head-gain scaling, not lower unit-direction tail sensitivity.	Both unit-JVP squared drift ratios exceed 1, but scaled and observed squared drift ratios are near 0.4766 [0.4442, 0.5114] and 0.596 [0.5623, 0.6318].	Spectral/polar unit directions are not always safer.

Table 3: Tail outcomes, margin-relevant drift, and head-gain readouts for the controlled one-step digits diagnostic. Small post-update effects are reported as changes from the pre-update tail state so that rounding of absolute CE or margin values does not hide the relevant comparison. The certificate fraction is computed only among tail samples with positive pre-update margin.

Readout	Value
Pre-update tail CE	11.74 [11.5, 11.99]
Tail CE increase, Fro/GD	−0.001154 [-0.004135, 0.001827]
Tail CE increase, spectral/polar	0.0002447 [-0.001859, 0.002348]
Tail CE increase difference, spectral–Fro	0.001399 [0.0002379, 0.002559]
Pre-update mean margin	−10.79 [-11.06, -10.52]
Margin drop, Fro/GD	0.001372 [-0.002944, 0.005688]
Margin drop, spectral/polar	0.003264 [0.0004973, 0.006032]
Margin-drop difference, spectral–Fro	0.001892 [9.142e-05, 0.003693]
Pre-update tail accuracy	0.1908 [0.1873, 0.1942]
Accuracy drop, Fro/GD	−0.00025 [-0.00074, 0.00024]
Accuracy drop, spectral/polar	−0.00025 [-0.00074, 0.00024]
Accuracy-drop difference, spectral–Fro	0 [0, 0]
Full tail-example logit drift, spectral/Fro	0.5501 [0.5101, 0.5931]
Centered-logit drift, spectral/Fro	0.5493 [0.5093, 0.5923]
True-class logit delta, spectral/Fro	1.68 [1.16, 2.431]
Top-competitor logit delta, spectral/Fro	0.6358 [0.5705, 0.7086]
Margin delta, spectral/Fro	0.702 [0.6131, 0.8037]
Actual head-gain relative error, Fro; spectral	0.02533 [0.02052, 0.03013]; 0.01662 [0.0148, 0.01845]
Head-alignment denominator, spectral/Fro	2.083 [1.991, 2.178]
Matched update Frobenius norm, spectral/Fro	3.112 [2.976, 3.255]
Matched update operator norm, spectral/Fro	0.5543 [0.5395, 0.5695]
Positive-margin fraction	0.1908
Certified unchanged-prediction fraction, Fro; spectral	0.9985; 0.9985
All-tail prediction changed, Fro; spectral	0.00625; 0.00575
Positive-margin prediction changed, Fro; spectral	0; 0

Appendix E gives the sufficient margin certificate used in Table 3. Only 0.1908 of tail examples have positive pre-update margin, with 95% CI [0.1873, 0.1942]. Among these examples, the certified unchanged-prediction fractions are 0.9985 for Fro/GD and 0.9985 for spectral/polar. The positive-margin prediction-change rate is zero for both directions, so the certificate check should be read as a consistency check on unchanged predictions among already-positive-margin examples, not as evidence of improved tail accuracy.

Table 4: Head-to-tail empirical evidence summary. Except for the practical imbalanced-training and CIFAR-100-LT final-training rows, which report actual training outcomes, ratios are spectral/polar or Muon-style directions divided by Frobenius/GD. Values below 1 mean lower perturbation of logits on held-out tail examples.

Experiment	Main tail-drift result	Key caveat
Synthetic positive boundary	0.3403 [0.3403, 0.3403]	$\mathrm{nrank}(G_H) = 8.61 > \mathrm{ssrank}(B_T, A_T) = 1.00$
Synthetic negative boundary	7.208 [7.208, 7.208]	$\mathrm{nrank}(G_H) = 1.39 < \mathrm{ssrank}(B_T, A_T) = 10.0$
Long-tailed digits one-step	0.5501 [0.5101, 0.5931]	tail loss diff = 0.001399 [2.38e-04, 0.002559]
CIFAR-100-LT ResNet18 one-step	0.5611 [0.5224, 0.6026]	tail loss diff = -4.21e-04 [-5.92e-04, -2.49e-04]; rho=0.002 ratio 0.7761 [0.7585, 0.7941]; accuracy diff CI crosses 0
CIFAR-100-LT ResNet18 checkpoint sweep	worst drift 0.5611 [0.5224, 0.6026] at 1.00e+03 steps	best pre-update tail accuracy 0.068 at 1.00e+03 steps; still weak-tail-predictor evidence
CIFAR-100-LT ResNet18 final-layer condition	worst drift 0.2391 [0.2201, 0.2597] at 5.00e+02 steps	weakest mean nrank/srank score 6.566; final-layer-only downstream-aware check
CIFAR-100 ResNet18 tail-quality control	worst drift 0.7292 [0.6923, 0.768]	best pre-update tail accuracy 0.3739 [0.3454, 0.4024]; tail-rich checkpoint control
CIFAR-100-LT ResNet18 imbalance sweep	worst drift 0.6075 [0.3943, 0.936] at tail train/class 1.00e+02	best pre-update tail accuracy 0.3297 [0.2954, 0.3639] at tail train/class 3.00e+02; tail-loss evidence is mixed
CIFAR-100-LT ResNet18 all-layer JVP	observed 0.2011 [0.1845, 0.2192]; scaled JVP 0.065 [0.06008, 0.07031]	21/21 layer CI upper endpoints below 1; tail-rich local JVP diagnostic
CIFAR-100-LT ResNet18 checkpoint-transfer JVP	scaled-JVP threshold accuracy 1; held-out Spearman -0.3203 [-0.3562, -0.2845]	positive controls: observed 0.981 [0.9739, 0.988], early-layer 0.9126 [0.9029, 0.9222]; 6 directed pairs
CIFAR-100-LT ResNet18 checkpoint-transfer residuals	observed residual Spearman 0.9403 [0.9216, 0.959]; scaled-JVP residual -0.4872 [-0.5373, -0.4372]	gradient-rank residual 0.3035 [0.2591, 0.3478]; source-fit early-layer residualization
CIFAR-100-LT ResNet18 candidate condition-score audit	early-layer prior Spearman 0.9126 [0.9029, 0.9222]; best simple composite 0.8656 [0.8578, 0.8734]	scaled-JVP raw/residual -0.3203 [-0.3562, -0.2845] / -0.4872 [-0.5373, -0.4372]; negative score-selection guardrail
CIFAR-100-LT ResNet18 standard reporting	many/medium/few balanced acc. 0.3665 [0.3489, 0.3841] / 0.1036 [0.09138, 0.1158] / 0.0129 [0.009351, 0.01645]	IF=100 AdamW ResNet18 reporting baseline; no augmentation, tuning, or Muon comparison
CIFAR-100-LT ResNet18 recipe pilot	SGD-aug all/few balanced acc. 0.4105 [0.4044, 0.4166] / 0.1047 [0.09389, 0.1156]	few diff vs AdamW-aug 0.0194 [0.005581, 0.03322]; class-balanced AdamW few diff -0.0114 [-0.0215, -0.001304]
CIFAR-100-LT ResNet18 NS-Muon final pilot	NS-Muon lr=10^{-4} all/few balanced acc. 0.1265 [0.1218, 0.1312] / 8.89e-04 [-6.53e-04, 0.002431]	AdamW-aug all/few 0.3613 [0.3595, 0.363] / 0.08856 [0.07566, 0.1014]; all/few diff -0.2348 [-0.2395, -0.23] / -0.08767 [-0.09948, -0.07585]; negative final-performance boundary
CIFAR-100-LT ResNet18 practical Muon bridge	AdamW-state $\mathrm{NS}(M_t)$ 0.8628 [0.8154, 0.913]; NS-Muon-state $\mathrm{NS}(M_t)$ 0.7247 [0.676, 0.777]	30 comparisons per state source/direction; local trajectory-state compatibility only
8-step head-only forgetting	0.6167 [0.5744, 0.6622]; area 0.7787 [0.7549, 0.8033]	tail loss/margin diff CI crosses 0
Muon-style compatibility: $\mathrm{polar}(M_t)$ / $\mathrm{NS}(M_t)$	0.8199 [0.6951, 0.9672] / 0.9116 [0.7696, 1.08]	$\mathrm{NS}(M_t)$ CI crosses 1; selected-state check only
Practical-Muon trajectory compatibility	0.7292 [0.6891, 0.7717] / 0.8019 [0.7583, 0.848]	120 state-step comparisons; still local matched-head-gain
Practical imbalanced training	train 0.6468 [0.604, 0.6926]; tail loss 0.8549 [0.8319, 0.8786]	tail accuracy diff = 0 [0, 0]; fixed lightweight hyperparameters
Layer 1 scaled JVP / observed	0.4766 [0.4442, 0.5114] / 0.4767 [0.4443, 0.5115]	unit-JVP squared drift ratio above one before head-gain scaling
Layer 2 scaled JVP / observed	0.596 [0.5623, 0.6318] / 0.596 [0.5623, 0.6318]	unit-JVP squared drift ratio above one before head-gain scaling

Table 5: Local linearization error for the controlled long-tailed digits fixed-checkpoint diagnostic. The relative error is $\|F_T(\theta + \Delta) - F_T(\theta) - J_T\Delta\| \,/\, \|J_T\Delta\|$ after matched-head-gain scaling.

Direction	Relative error	$\|J_T\Delta\|_F$	Residual norm
Fro/GD	1.42e-04 [9.51e-05, 1.90e-04]	0.7796 [0.6733, 0.8858]	1.24e-04 [7.34e-05, 1.75e-04]
$\mathrm{polar}(G_t)$	0.001348 [6.96e-04, 0.001999]	0.572 [0.5017, 0.6424]	8.62e-04 [4.03e-04, 0.001322]
$\mathrm{polar}(M_t)$	0.001507 [9.64e-04, 0.00205]	0.7258 [0.5687, 0.883]	0.001201 [6.33e-04, 0.001769]
$\mathrm{NS}(M_t)$	0.001066 [6.88e-04, 0.001444]	0.7721 [0.5925, 0.9517]	9.00e-04 [5.08e-04, 0.001292]

K Auxiliary Diagnostic Figures

The synthetic alignment ablation keeps the singular values from the boundary diagnostic but randomizes the singular vectors of the head gradient, tail activation, and downstream tail map. Figure 4 shows that the positive-spectrum bound ratio remains below 1, while the realized drift ratio becomes mixed under random alignment. This is a direct caveat on interpreting $\operatorname{nrank}(G_H) > \operatorname{ssrank}(B_T, A_T)$: the condition orders worst-case sensitivity bounds, whereas realized polar drift also depends on singular-vector alignment.

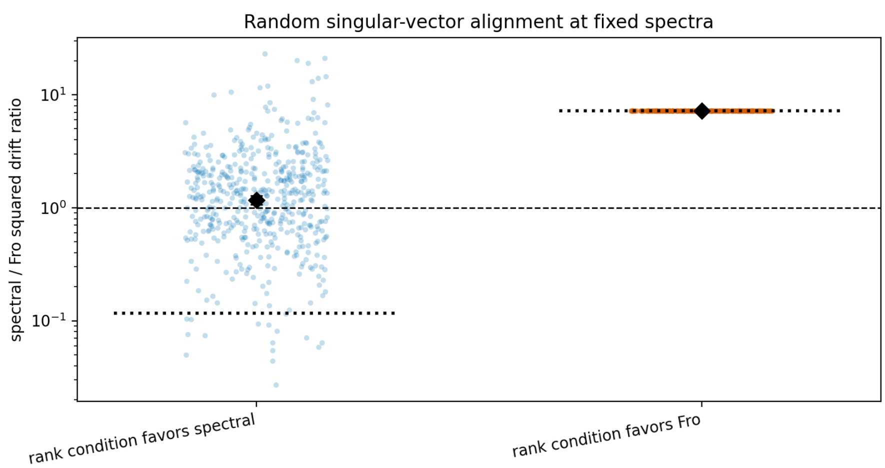

Figure 4: Synthetic singular-vector alignment ablation. Each point randomizes the singular-vector alignment of G_H, A_T, B_T while preserving the singular values used in the synthetic boundary diagnostic. The dotted horizontal markers show the theory ratio $\operatorname{ssrank}(B_T, A_T)/\operatorname{nrank}(G_H)$; black diamonds report the geomean realized squared drift ratio with 95% intervals. In the positive-spectrum setting, the bound favors spectral geometry, but randomized alignment makes the realized drift ratio mixed.

The fixed-checkpoint Muon-style compatibility diagnostic tests whether replacing $\operatorname{polar}(G_t)$ by Muon-style state gives a compatible local drift signal. Figure 5 shows that $\operatorname{polar}(M_t)$ has a matched-head-gain squared tail-example logit drift ratio below 1 at the same checkpoint, while the finite-step Newton–Schulz approximation has a squared drift-ratio interval crossing 1. Accordingly, this result provides selected-state compatibility evidence from $\operatorname{polar}(G_t)$ to Muon-style directions, rather than an optimizer-level performance claim.

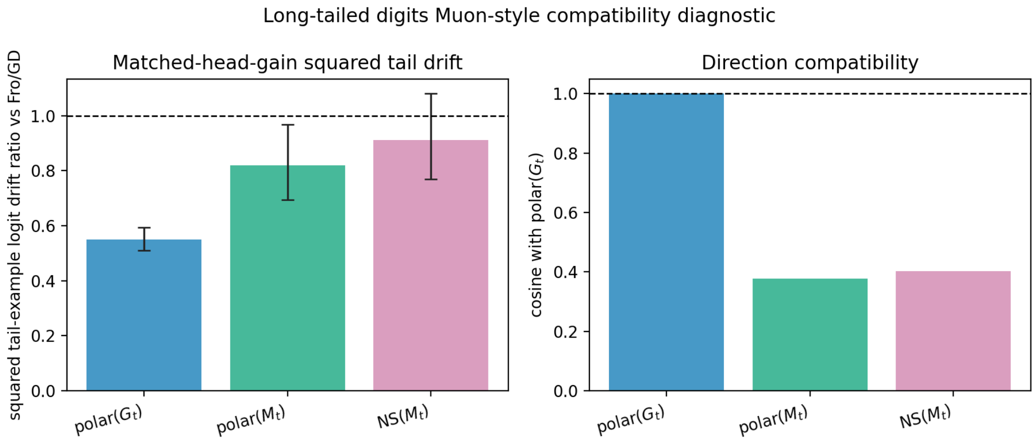

Figure 5: Fixed-checkpoint Muon-style direction compatibility diagnostic on controlled long-tailed digits. The left panel compares the matched-head-gain squared tail-example logit drift ratio of several directions relative to Frobenius/GD. The right panel reports their cosine with the ideal $\mathrm{polar}(G_t)$. Momentum polar has a squared drift ratio below 1, while the finite Newton–Schulz approximation has a squared drift-ratio interval crossing 1.

The state-source control asks whether the short-trajectory Muon-style signal depends on evaluating Muon-style directions only at states generated by the same NS-Muon-style family. Figure 6 repeats the matched-head-gain diagnostic at states generated by either NS-Muon-style updates or Fro/GD-style updates. The Fro/GD-state control gives $\mathrm{polar}(M_t)$ squared drift ratio 0.7255 [0.686, 0.7672] and NS(M_t) ratio 0.7973 [0.7546, 0.8425] across 120 comparisons, reducing but not eliminating state-selection risk.

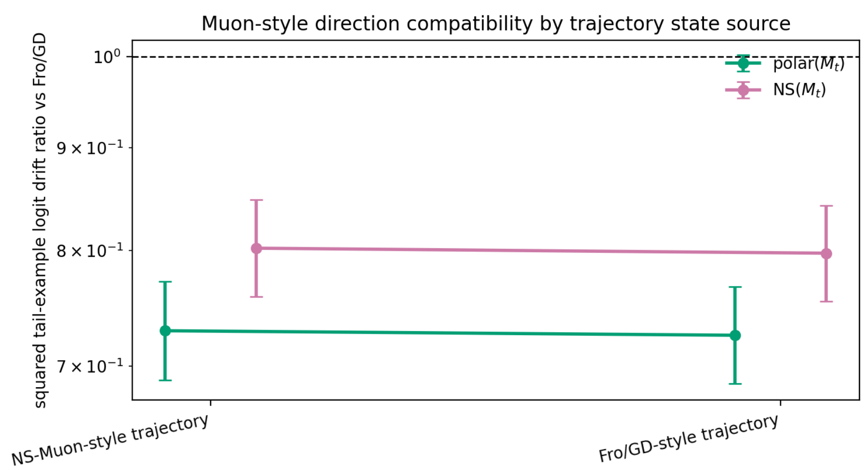

Figure 6: Muon-style direction compatibility by trajectory state source. The diagnostic is evaluated on sampled states from the original short NS-Muon-style trajectory and from a Fro/GD-style trajectory with matched seeds and batches. Values below 1 mean lower held-out squared tail-example logit drift than the Fro/GD direction at the same state after matching first-order head gain.

The imbalance ablation repeats the one-step matched-head-gain diagnostic while varying the number of tail training examples per class. Figure 7 shows that the spectral/Frobenius squared drift-ratio confidence intervals remain below 1 across the tested 100:80, 100:40, 100:20, and 100:10 splits. The strongest-imbalance case has squared drift ratio 0.5881 [0.5437, 0.6361], but its positive-margin tail fraction is 0 [0, 0], so margin-certificate conclusions are not meaningful there.

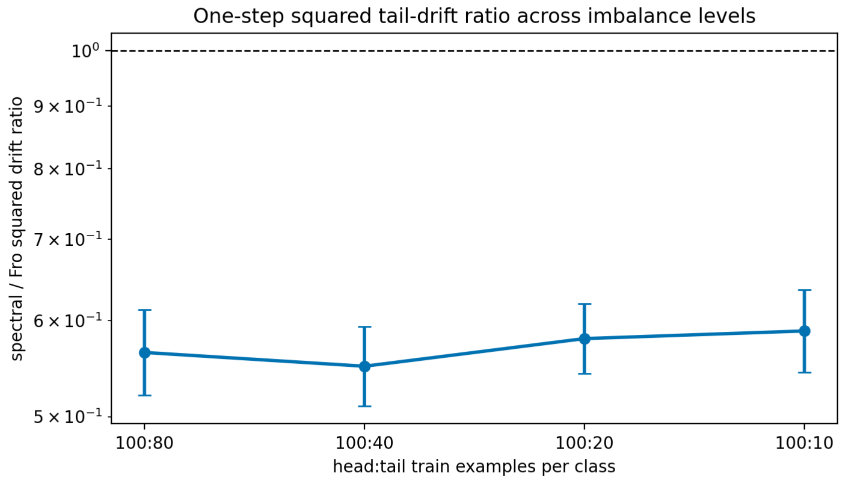

Figure 7: Imbalance ablation for the controlled long-tailed digits one-step diagnostic. The head split remains fixed at 100 training examples per head class, while tail training examples per class vary. Values below 1 mean lower held-out squared tail-example logit drift for spectral/polar after matching first-order head gain.

The checkpoint sweep repeats the same matched-head-gain diagnostic at multiple warmup states instead of reporting only the default 80-step checkpoint. Figure 8 shows that the full tail-example squared logit drift ratio remains below 1 across 5 checkpoints; the worst upper endpoint is 0.7071 at 160 warmup steps. The squared margin-delta ratio is less uniform: its worst interval is 1.081 [0.8477, 1.379] at 20 warmup steps, so the checkpoint sweep supports the logit drift claim more directly than a margin-delta claim. This check reduces single-checkpoint state-selection risk for the drift readout, but it does not yet show preservation of a high-quality tail predictor: across the tested checkpoints, pre-update tail accuracy spans only 0 to 0.192, and the positive-margin tail fraction spans 0 to 0.192. A stronger tail-knowledge preservation claim therefore requires checkpoints or benchmarks where the tail function is already useful, not only less perturbed.

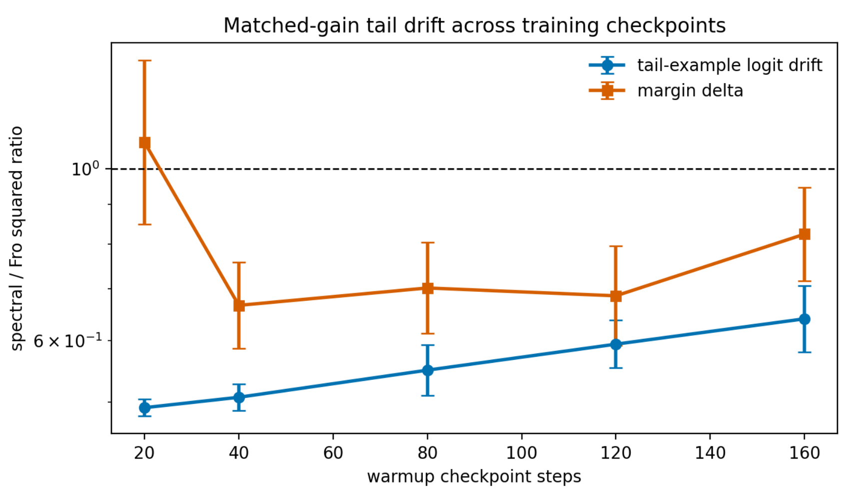

Figure 8: Checkpoint sweep for the controlled long-tailed digits one-step diagnostic. Each point repeats the matched-head-gain Frobenius/GD versus spectral/polar comparison at a different warmup checkpoint. Values below 1 mean lower held-out squared tail-example logit drift or lower squared margin delta for spectral/polar after matching first-order head gain.

The class-partition sweep changes which digit classes serve as head and tail classes. Figure 9 shows that the full and centered tail-example squared logit drift ratios remain below 1 across 5 balanced partitions. The weakest full tail-example squared logit drift case is `mixed_c`, with ratio 0.6135 [0.5762, 0.6532]; the weakest centered squared logit drift case is `mixed_c`, with ratio 0.6124 [0.5751, 0.652]. This reduces the risk that the result is an artifact of using digits 0–4 as head classes and 5–9 as tail classes, while still remaining within the same small dataset.

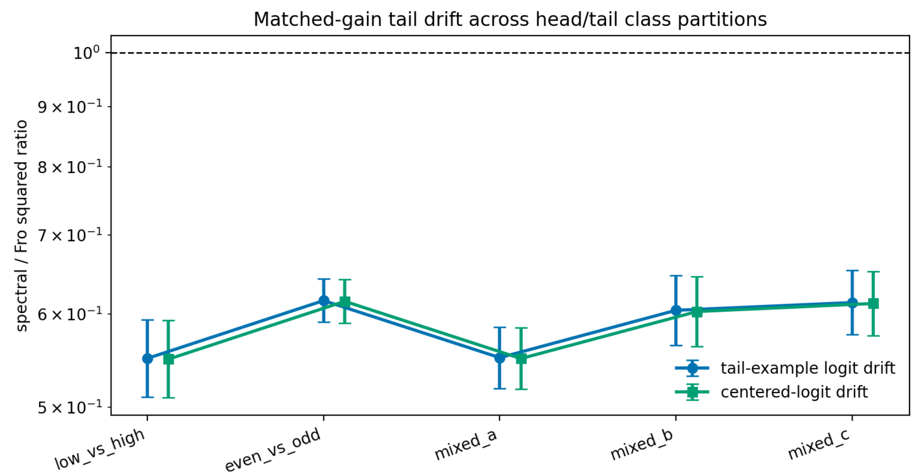

Figure 9: Class-partition sweep for the controlled long-tailed digits one-step diagnostic. Each point repeats the matched-head-gain comparison after changing which five digit classes are head classes and which five are tail classes. Values below 1 mean lower held-out squared tail-example logit drift for spectral/polar after matching first-order head gain.

The matched-gain rho sweep changes the target first-order head gain ρ while keeping the same controlled digits checkpoint distribution. Figure 10 shows that the full and centered tail-example squared logit drift ratios remain below 1 across 5 tested fractions. The weakest full tail-example squared logit drift case is $\rho/L_H = 0.005$, with ratio 0.5501 [0.5101, 0.5932]; the weakest centered squared logit drift case is $\rho/L_H = 0.005$, with ratio 0.5493 [0.5093, 0.5924]. At the largest tested scale, the one-step approximation is less exact: the largest actual-head-gain relative-error upper endpoint across both geometries is 0.1176 at $\rho/L_H = 0.08$. Thus the sweep supports the drift readout across local step scales, while preserving the paper's first-order scope.

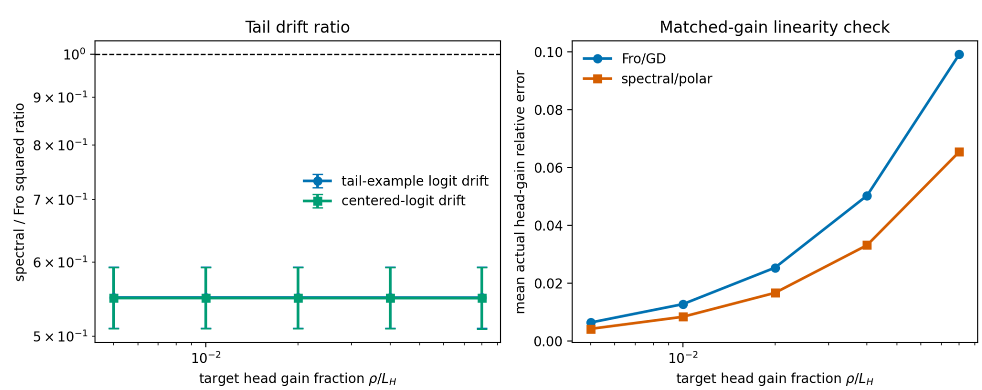

Figure 10: Target head-gain fraction sweep for the controlled long-tailed digits one-step diagnostic. The left panel reports full and centered squared tail-example logit drift ratios after matching first-order head gain. The right panel reports the actual head-gain relative error, which increases with ρ/L_H and marks the finite-step cost of leaving the most local regime.

The practical imbalanced-training diagnostic is a non-matched-gain sanity check reported for context, not evidence for the main theorem. Figure 11 compares Adam and a finite-Newton–Schulz Muon-style update under one controlled imbalanced-training protocol. The experiment is not a complete optimizer benchmark because tail accuracy does not improve stably, hyperparameters are lightweight, and standard long-tailed visual benchmarks are not covered. The final train-loss ratio is 0.6468 [0.604, 0.6926], and the tail-margin difference is 1.955 [1.628, 2.281]. The learning-rate sweep is a caveat: the largest Muon learning rate raises the tail-loss ratio to 2.049, indicating sensitivity to step size rather than a monotone improvement from larger Muon updates.

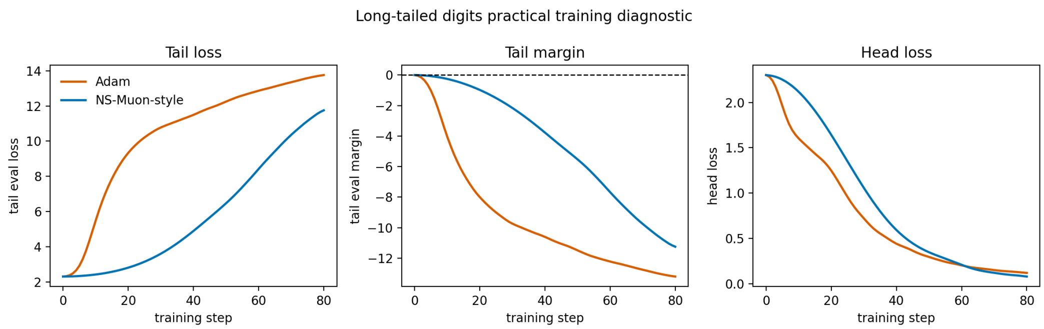

Figure 11: Controlled long-tailed digits practical training diagnostic. Adam and the finite-Newton–Schulz Muon-style update start from the same initialization and use the same noisy mini-batch schedule. In this controlled imbalanced setting, the Muon-style run reports lower tail loss, a higher measured tail-margin reading, and lower late-stage head loss; this protocol does not constitute a complete optimizer benchmark.

The forgetting probe repeats head-only updates before rechecking the tail set. Figure 12 reports lower tail-example logit drift over a short head-only horizon, but the intervals for tail loss and margin are still insufficient to support a performance claim. The final tail-margin drop difference is -0.003658 [-0.009636, 0.00232]. We therefore treat the forgetting probe as drift evidence rather than a tail-performance claim.

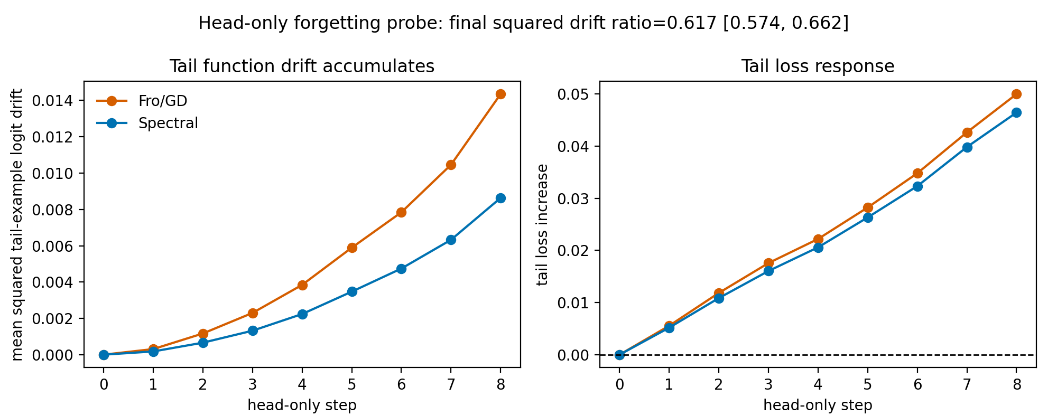

Figure 12: Tail forgetting probe under consecutive head-only steps. The left panel shows that spectral/polar squared tail-example logit drift remains below Frobenius/GD across the short horizon. The right panel shows a weaker tail-loss-increase difference, which does not establish a stable performance claim.

Figure 13 reports the layerwise drift diagnostic discussed in the main text.

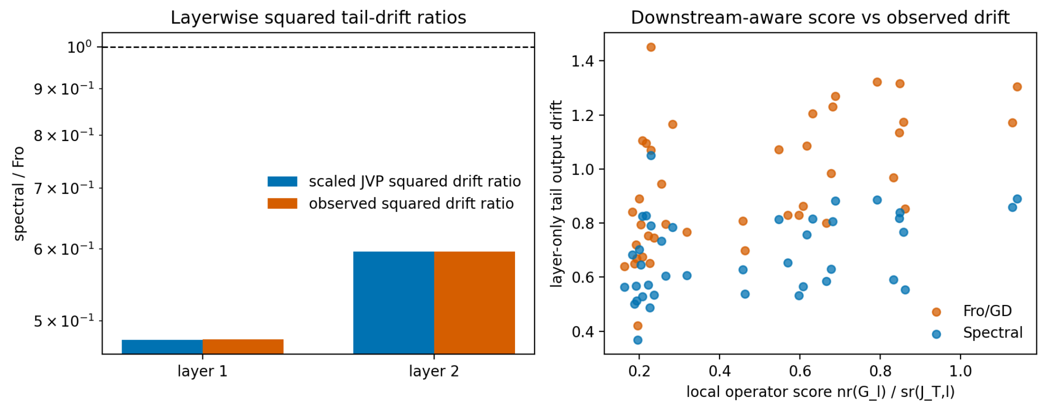

Figure 13: Layerwise tail-drift diagnostic. Both layers have unit-direction JVP squared drift ratios above 1, showing that spectral/polar unit directions are not always less tail-perturbing. After matched-head-gain scaling, however, both scaled JVP and observed squared drift ratios fall below the Frobenius/GD baseline. The right panel uses the exact frozen-gate local operator score; for the final layer this is supplemented in the text by the exact sandwich quantity $\mathrm{ssrank}(B_{T,2}, A_{T,2})$.

Figure 14 reports the all-layer ResNet finite-difference JVP diagnostic discussed in the ResNet subsection.

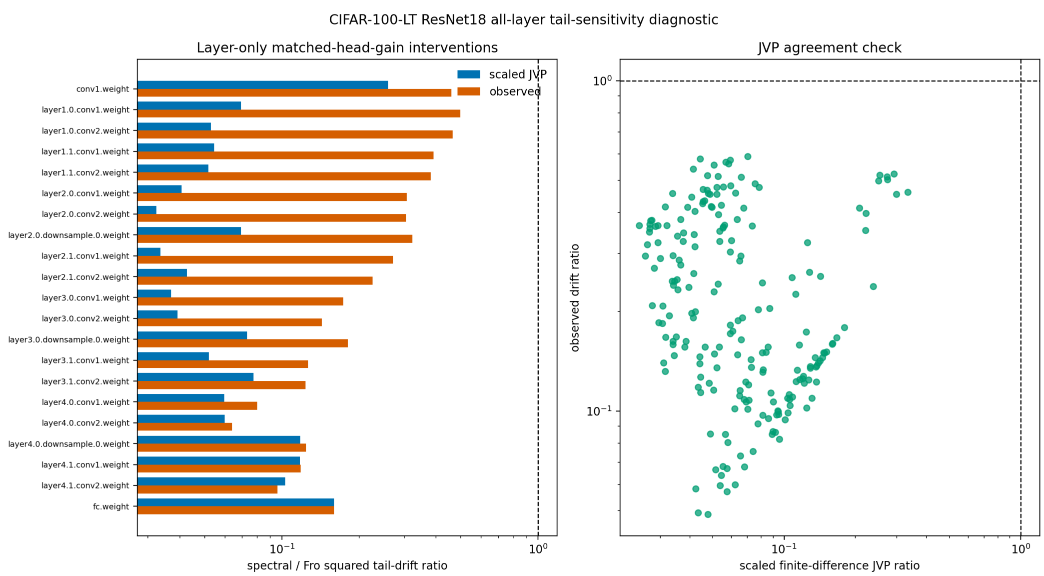

Figure 14: CIFAR-100-LT ResNet18 all-layer JVP tail-quality diagnostic. The left panel reports matched-head-gain scaled-JVP and observed squared drift ratios for every Conv/Linear matrix weight at the tail-rich checkpoint; values below 1 favor spectral/polar. The right panel compares finite-difference scaled-JVP ratios to observed layer-only drift ratios across seed/layer pairs. This is local mechanism evidence, not a practical optimizer benchmark.

Figure 15 turns the all-layer ResNet JVP diagnostic into a held-out checkpoint-transfer check. Across 3 tail-rich checkpoints and 63 layer/checkpoint summaries, source-observed drift and an early-layer architecture prior have held-out checkpoint layer-risk Spearman 0.981 [0.9739, 0.988] and 0.9126 [0.9029, 0.9222], respectively. After source-fit residualization against the early-layer prior, observed residual risk still transfers with Spearman 0.9403 [0.9216, 0.959], whereas the scaled-JVP residual has Spearman -0.4872 [-0.5373, -0.4372]. The scaled-JVP predictor keeps the directional below-one threshold accuracy at 1, but its held-out checkpoint layer-risk Spearman is -0.3203 [-0.3562, -0.2845]. We therefore treat this as a sharper boundary result: held-out layer-risk ordering is predictable in this ResNet family, including depth-adjusted residual structure, but the current local JVP readout explains matched-head-gain directionality rather than cross-checkpoint layer ranking.

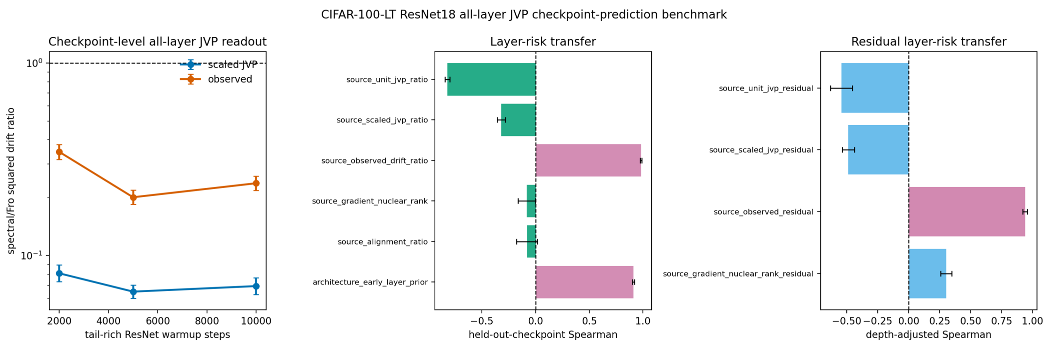

Figure 15: CIFAR-100-LT ResNet18 all-layer JVP checkpoint-transfer benchmark. The left panel checks whether the all-layer directional drift ratios remain below the Frobenius baseline across tail-rich checkpoints. The middle panel reports held-out checkpoint Spearman correlations for raw source-checkpoint layer scores against target-checkpoint observed layer drift. The right panel repeats the transfer test after source-fit residualization against early-layer structure. Positive controls show that layer-risk ordering is stable, while the negative scaled-JVP raw and residual rankings mark a current predictive-boundary limitation.

Figure 16 audits simple source-only candidate scores on the same checkpoint-transfer benchmark. The source-observed positive control and early-layer prior have Spearman 0.981 [0.9739, 0.988] and 0.9126 [0.9029, 0.9222], while the best simple condition composite in this audit, early-minus-scaled-JVP, reaches 0.8656 [0.8578, 0.8734], below the architecture prior. After source-fit depth residualization, observed residual Spearman remains 0.9403 [0.9216, 0.959], but scaled-JVP residual Spearman is -0.4872 [-0.5373, -0.4372]. We use this as a guardrail against post-hoc score selection: the current obvious score family does not solve the predictive condition problem.

A later score-axis ablation keeps this claim boundary explicit. On the spent v4 CIFAR-10 mixed final split, the direction-axis residual Spearman is positive, 0.611 $[0.5886, 0.6334]$, while the raw Frobenius-amplitude axis reverses, -0.6766 $[-0.6903, -0.663]$, and the v4 amplitude-minus-direction scalar also reverses, -0.6937 $[-0.7129, -0.6744]$. Thus a below-one direction guardrail is not a residual layer-risk ranking claim. The v5 validation-only split freezes a transport-normalized candidate with residual Spearman 0.645 $[0.5898, 0.7002]$, above the early-layer baseline 0.2996 $[0.2408, 0.3583]$, but this remains validation evidence: the unspent ResNeXt50-32x4d architecture final and CIFAR-10 cross-partition data final must both pass before any predictive-condition claim is supportable.

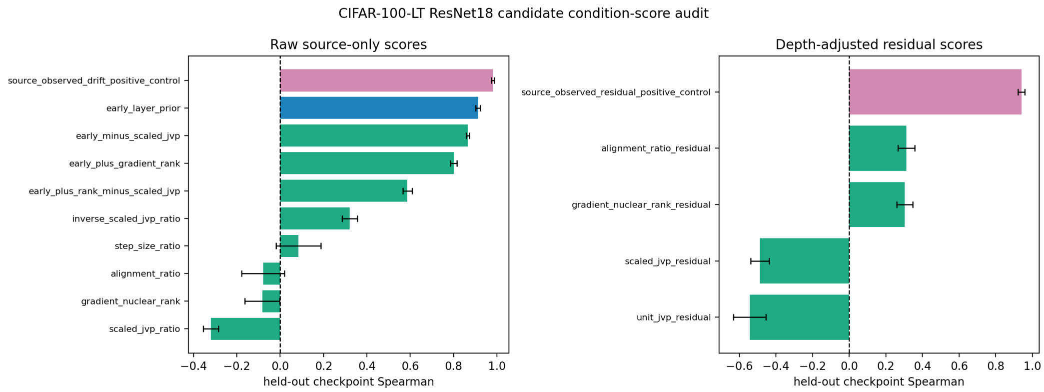

Figure 16: CIFAR-100-LT ResNet18 candidate condition-score audit. The left panel evaluates raw source-only scores for held-out checkpoint layer-risk ranking. The right panel repeats the comparison after source-fit early-layer residualization. Simple head-rank, JVP, and depth-composite scores do not beat the architecture prior, and scaled-JVP remains inverted in the residual test.

Figure 17 sweeps explicit tail-count settings for the same CIFAR-100-LT ResNet18 local diagnostic. Across 4 settings, the weakest squared drift-ratio interval remains below the Frobenius baseline: 0.6075 [0.3943, 0.936] at tail count 100. The best pre-update tail accuracy is 0.3297 [0.2954, 0.3639] at tail count 300, where the drift ratio is 0.3898 [0.3143, 0.4835]. Tail-loss differences are not uniformly favorable, so the figure is a tail-frequency robustness check for the local function-drift claim.

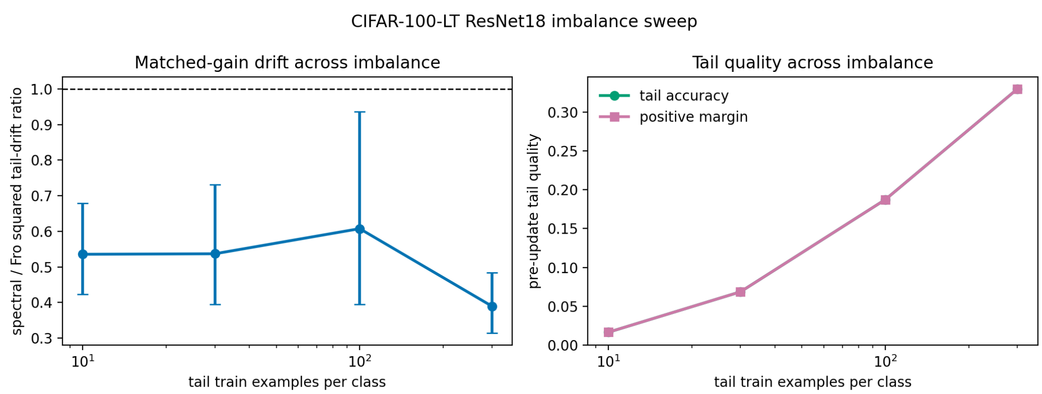

Figure 17: CIFAR-100-LT ResNet18 tail-count imbalance sweep. The diagnostic holds the head count fixed at 300 examples per class and varies the tail count across 10, 30, 100, and 300 while matching the same first-order head gain. Values below 1 mean lower held-out squared tail-example logit drift for spectral/polar than Frobenius/GD.

Figure 18 adds a standard CIFAR-100-LT reporting surface for final classification metrics. Under an IF=100 exponential split, AdamW ResNet18 over 10 seeds gives many/medium/few balanced accuracy 0.3665 [0.3489, 0.3841], 0.1036 [0.09138, 0.1158], and 0.0129 [0.009351, 0.01645]. This appendix result separates the paper's local matched-head-gain drift claims from benchmark-style classification readouts; it is not an augmented, tuned, or multi-optimizer comparison.

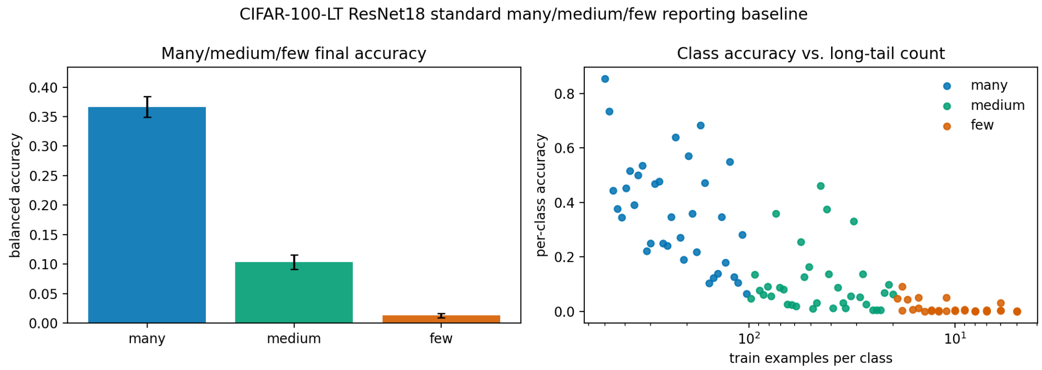

Figure 18: CIFAR-100-LT ResNet18 standard many/medium/few reporting baseline. The run uses an IF=100 exponential class-count split and reports final AdamW balanced accuracy by frequency group. It is a classification reporting surface for context, not evidence that the local drift mechanism improves final long-tail accuracy.

Figure 19 adds a small augmented recipe pilot. With random crop/flip augmentation over 5 seeds, AdamW-aug reaches all-group balanced accuracy 0.3598 [0.3571, 0.3625]. SGD-momentum-aug reaches 0.4105 [0.4044, 0.4166] overall and 0.1047 [0.09389, 0.1156] on few classes, with few-class paired difference 0.0194 [0.005581, 0.03322] versus AdamW-aug. The class-balanced AdamW loss is not a free improvement in this pilot: its few-class difference is -0.0114 [-0.0215, -0.001304]. This result improves benchmark context but still falls short of a full tuned long-tail optimizer comparison.

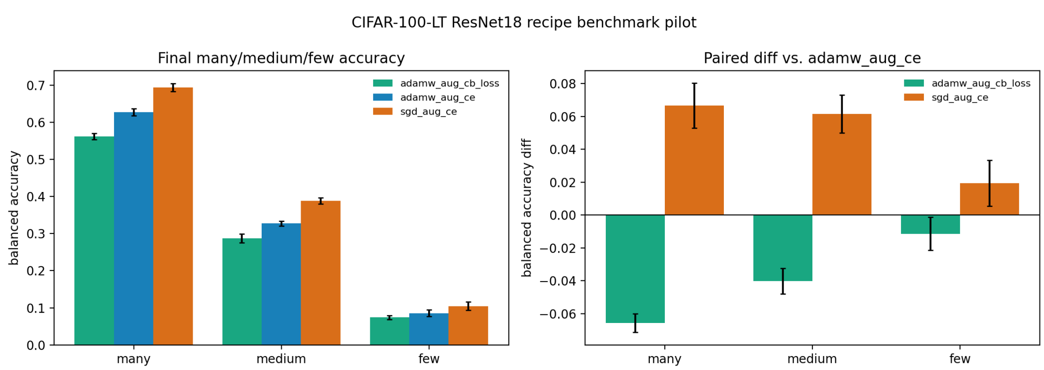

Figure 19: CIFAR-100-LT ResNet18 augmented recipe benchmark pilot. The left panel reports final many/medium/few balanced accuracy for augmented AdamW, class-balanced AdamW, and SGD-momentum. The right panel reports paired balanced-accuracy differences relative to augmented AdamW. The pilot shows meaningful recipe sensitivity and supports keeping optimizer-performance claims separate from the local drift mechanism.

Figure 20 adds the direct NS-Muon final-training pilot. The lr 10^{-4} NS-Muon recipe reaches all/few balanced accuracy 0.1265 [0.1218, 0.1312] and 0.0008889 [-0.0006533, 0.002431], versus augmented AdamW at 0.3613 [0.3595, 0.363] and 0.08856 [0.07566, 0.1014]. The paired all/few differences are -0.2348 [-0.2395, -0.23] and -0.08767 [-0.09948, -0.07585].

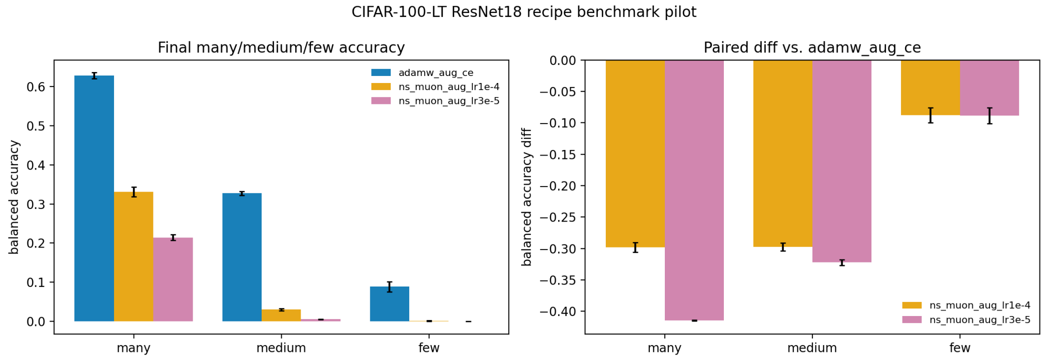

Figure 20: CIFAR-100-LT ResNet18 NS-Muon final-training benchmark pilot. The left panel reports final many/medium/few balanced accuracy for augmented AdamW and two finite-Newton-Schulz Muon-style matrix-weight training recipes. The right panel reports paired balanced-accuracy differences relative to augmented AdamW. This negative final-performance boundary is not a contradiction of the local Muon bridge; it shows that the tested practical NS-Muon recipe needs better schedules and tuning before any optimizer-performance claim.

Figure 21 reports the ResNet practical trajectory bridge discussed in the Muon-style compatibility subsection.

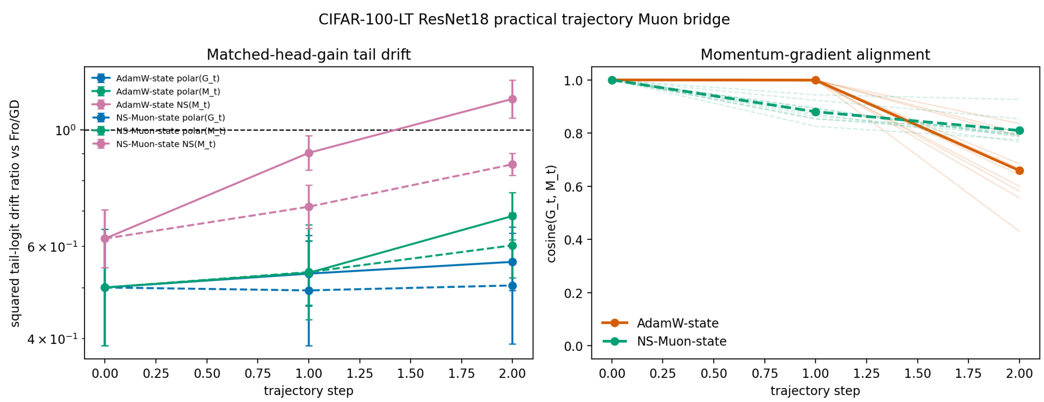

Figure 21: CIFAR-100-LT ResNet18 practical Muon trajectory bridge. The diagnostic starts from a tail-rich ResNet checkpoint, samples matrix-weight-only AdamW and finite-step NS-Muon trajectory states, and compares $\mathrm{polar}(G_t)$, $\mathrm{polar}(M_t)$, and $\mathrm{NS}(M_t)$ against Fro/GD after matching the same head-batch first-order gain. This is a local trajectory-state compatibility check, not a tuned long-tailed optimizer benchmark.